

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Automated evidence-grounded report

August 28, 2026

Abstract

A C++17/MPI cell-centred finite-volume solver was applied to all eight supplied unstructured CGNS cases using METIS partitions, second-order limited reconstruction, Rusanov convective flux, and Newtonian/Fourier viscous flux where applicable. 8 of 8 required cases completed with measured sanity checks passing; seven are steady and the Re200 cylinder is statistically periodic. The Re200 run reached a statistically periodic state with mean $C_D = 1.1399$, lift amplitude 0.33817, and $St = 0.16667$. Known accuracy and visualization limitations are stated explicitly.

1 Introduction

The benchmark covers three inviscid and three laminar NACA0012 cases at zero angle of attack, plus laminar circular-cylinder cases at Reynolds 20 and 200. It exercises mesh import, compressible-flow numerics, implicit steady and physical-time transient integration, distributed-memory execution, restart/output integrity, and traceable post-processing. Results below distinguish final steady samples from post-transient statistics.

2 Governing equations and nondimensionalization

The conservative state is $\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T$ and $\partial_t \mathbf{U} + \partial_x \mathbf{F}^i + \partial_y \mathbf{G}^i = \partial_x \mathbf{F}^v + \partial_y \mathbf{G}^v$. The inviscid fluxes are $\mathbf{F}^i = [\rho u, \rho u^2 + p, \rho uv, u(\rho E + p)]^T$ and $\mathbf{G}^i = [\rho v, \rho uv, \rho v^2 + p, v(\rho E + p)]^T$, with $p = (\gamma - 1)\rho e$, $E = e + (u^2 + v^2)/2$, and $a = \sqrt{\gamma p / \rho}$. For viscous cases the flux contains Newtonian stresses $\tau_{ij} = \mu(\partial_i u_j + \partial_j u_i - \frac{2}{3}\delta_{ij}\nabla \cdot \mathbf{u})$ and Fourier heat flux $q_i = -\mu c_p \partial_i T / Pr$. Constant viscosity is set by $Re = \rho_\infty U_\infty L / \mu$. The case reference density, velocity, and length nondimensionalize the solution; pressure uses $\rho_\infty U_\infty^2$, while $C_D = F_D / (q_\infty A)$, $C_L = F_L / (q_\infty A)$, and $C_m = M / (q_\infty AL)$ with $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$.

3 Meshes, boundary tags, and case inputs

Rank 0 reads the CGNS unstructured zone, polygon connectivity, and named boundary sections. It constructs consistently oriented faces from cell edges, pairs interior edges, assigns boundary-family tags, and computes polygon area/centroid plus face midpoint, outward unit normal, and length. Boundary mappings are checked against the mesh before marching. Global sizes and supplied parameters are reproduced below.

Case	mesh	cells	faces	Mach	Re	boundary tags
naca0012_m015_inviscid	NACA0012_H2.cgns	20816	36498	0.15	–	bc-2:farfield, bc-4:slip_wall
naca0012_m080_inviscid	NACA0012_H2.cgns	20816	36498	0.8	–	bc-2:farfield, bc-4:slip_wall
naca0012_m200_inviscid	NACA0012_H2.cgns	20816	36498	2.0	–	bc-2:farfield, bc-4:slip_wall
naca0012_m015_laminar_re5000	NACA0012_H2.cgns	20816	36498	0.15	5000.0	bc-2:farfield, bc-4:no_slip_adiabatic_wall
naca0012_m080_laminar_re5000	NACA0012_H2.cgns	20816	36498	0.8	5000.0	bc-2:farfield, bc-4:no_slip_adiabatic_wall
naca0012_m200_laminar_re5000	NACA0012_H2.cgns	20816	36498	2.0	5000.0	bc-2:farfield, bc-4:no_slip_adiabatic_wall

cylinder_m010_laminar_re20	CylinderB1.cgns	10185	20180	0.1	20.0	WALL:no_slip-adiabatic.wall, FAR:farfield
cylinder_m010_laminar_re200	CylinderB1.cgns	10185	20180	0.1	200.0	WALL:no_slip-adiabatic.wall, FAR:farfield

4 Spatial discretization

For a cell K of area V_K , the cell-centred finite-volume residual is $R_K = \sum_{f \in \partial K} (\hat{F}_f^i - \hat{F}_f^v) |f|$, with each stored normal directed out of the left cell; the same interior-face flux is added to the left residual and subtracted from the right residual. The semi-discrete equation is $V_K dU_K/dt + R_K = 0$. Boundary faces use a reconstructed interior state and a boundary/ghost state described below.

4.1 Second-order reconstruction, limiter, and positivity

Primitive gradients solve the weighted least-squares normal equations $A_K \nabla q_K = b_K$ from neighbor-centroid offsets, including exchanged ghost values and boundary values. Face states use $q_f = q_K + \alpha_K \nabla q_K \cdot (x_f - x_K)$. A Barth–Jespersen limiter chooses $0 \leq \alpha_K \leq 1$ so every extrapolated primitive remains inside local neighbor bounds. Production steady runs ramp the reconstruction factor from zero during startup to one, so the submitted terminal solutions use the full limited linear reconstruction. If reconstructed density or pressure is non-positive, that face side falls back to the admissible cell-centre state; conservative updates are likewise line-searched toward the old state until density and pressure are positive.

4.2 Inviscid and viscous fluxes

The convective numerical flux is Rusanov/local Lax–Friedrichs, $\hat{F}^i = \frac{1}{2}[F_n(U_L) + F_n(U_R)] - \frac{1}{2}s_{\max}(U_R - U_L)$, where $s_{\max} = \max(|u_{n,L}| + a_L, |u_{n,R}| + a_R)$ (times the recorded dissipation scale). No separate entropy fix is used. Laminar face velocity and temperature gradients are averaged from neighboring least-squares gradients. They form Newtonian stress and Fourier heat flux; at no-slip walls the normal velocity gradient is imposed from the cell-to-wall distance and the adiabatic temperature-normal gradient is zero. Pressure force and tangential wall shear are accumulated separately, so skin friction excludes viscous normal traction.

5 Boundary conditions

Farfield faces use a characteristic-style boundary state: incoming information is drawn from the prescribed freestream and outgoing information from the interior. A slip wall reflects normal velocity in the ghost state while retaining tangential velocity; its direct wall state has zero normal velocity. A no-slip adiabatic wall reflects both velocity components so the wall-interpolated velocity is zero, retains wall temperature for zero normal heat flux, and applies direct pressure plus tangential-shear wall fluxes. Consequently **surface.csv** contains boundary values, not adjacent cell-centre values; the measured slip/no-slip gates are reported in the sanity section.

Case	inviscid flux	viscous flux	reconstruction	limiter	positivity	implicit solver
naca0012_m015_inviscid	Rusanov	disabled	LS linear	Barth–Jespersen	ρ, p fallback	Schwarz LU–SGS
naca0012_m080_inviscid	Rusanov	disabled	LS linear	Barth–Jespersen	ρ, p fallback	Schwarz LU–SGS
naca0012_m200_inviscid	Rusanov	disabled	LS linear	Barth–Jespersen	ρ, p fallback	Schwarz LU–SGS
naca0012_m015_laminar_re5000	Rusanov	Newtonian/Fourier	LS linear	Barth–Jespersen	ρ, p fallback	Schwarz LU–SGS
naca0012_m080_laminar_re5000	Rusanov	Newtonian/Fourier	LS linear	Barth–Jespersen	ρ, p fallback	Schwarz LU–SGS
naca0012_m200_laminar_re5000	Rusanov	Newtonian/Fourier	LS linear	Barth–Jespersen	ρ, p fallback	Schwarz LU–SGS
cylinder_m010_laminar_re20	Rusanov	Newtonian/Fourier	LS linear	Barth–Jespersen	ρ, p fallback	Schwarz LU–SGS
cylinder_m010_laminar_re200	Rusanov	Newtonian/Fourier	LS linear	Barth–Jespersen	ρ, p fallback	Schwarz LU–SGS

6 Implicit and transient time integration

Steady marching uses a local pseudo-time step $\Delta\tau_K = \text{CFL } V_K / \lambda_K$. The scalar diagonal contains convective spectral radii $(|u_n| + a)|f|$, a viscous contribution proportional to $\mu|f|/(\rho d)$, and any physical-time coefficient. CFL is ramped geometrically from the supplied initial to maximum value. Each nonlinear iteration applies forward and backward local cell sweeps to the scalar spectral Jacobian; exchanged ghost increments make this a distributed additive-Schwarz LU-SGS relaxation. Global L_2/L_∞ residuals and forces use MPI reductions, and steady acceptance requires the recorded residual decrease plus a stable terminal force plateau. For Re200 the true BDF2 inner loop advances exactly $\Delta t = 0.01$ to $t = 300$. Backward Euler starts the first step; later steps solve $V_K(3U^{n+1} - 4U^n + U^{n-1})/(\Delta t) + R_K(U^{n+1}) = 0$. Both U^n and U^{n-1} remain frozen throughout every inner iteration, which evaluates the full spatial-plus-physical residual. History shifts only after the inner reduction target or permitted maximum is reached. This is a true physical-time outer/inner algorithm, not pseudo-time-only marching.

Case	max steps	CFL initial/max	ramp steps	inner min/max	target
naca0012_m015_inviscid	20000	1.0/100.0	2000	3/50	0.01
naca0012_m080_inviscid	30000	1.0/100.0	3000	3/50	0.01
naca0012_m200_inviscid	40000	0.2/50.0	5000	3/80	0.01
naca0012_m015_laminar_re5000	40000	1.0/100.0	5000	3/80	0.01
naca0012_m080_laminar_re5000	40000	0.5/80.0	5000	3/80	0.01
naca0012_m200_laminar_re5000	50000	0.2/50.0	7000	3/100	0.01
cylinder_m010_laminar_re20	30000	1.0/100.0	3000	3/80	0.01
cylinder_m010_laminar_re200	–	1.0/1.0	0	5/1000	0.001

Observed iteration statistics and physical-time controls are:

Case	observed inner min/max/mean	$\Delta t/t_f$	completed steps
naca0012_m015_inviscid	3/3/3	–/–	10000
naca0012_m080_inviscid	3/3/3	–/–	10000
naca0012_m200_inviscid	3/3/3	–/–	2120
naca0012_m015_laminar_re5000	3/3/3	–/–	10000
naca0012_m080_laminar_re5000	3/3/3	–/–	10000
naca0012_m200_laminar_re5000	3/3/3	–/–	5503
cylinder_m010_laminar_re20	3/3/3	–/–	10000
cylinder_m010_laminar_re200	110/1000/172.13	0.01/300.0	30000

7 Output, reproducibility, and run status

Configure with `cmake-Ssolver-Bsolver/build-DCMAKE_BUILD_TYPE=Release-DBUILD_TESTING=ON`, then run `cmake--buildsolver/build-j`. CMake locates MPI, CGNS, METIS, and nlohmann-json and records their resolved paths in the build cache. Regenerate evidence with `.venv/bin/pythontools/generate_report.py--resultsresults--reportreport` and the supplied `--case-inputs` directory. Each output contains metadata/status JSON, partition diagnostics, residual/force/surface CSV files, the PVTU master plus rank VTU pieces, rank restart pieces, and stdout. The machine-readable run, figure, rank-comparison, spectrum, and sanity manifests accompany this report. Exact 32-rank production commands are retained in the run manifest and printed below.

Case	ranks	steps	time	residual	orders	wall s	status
naca0012_m015_inviscid	32	10000	0		2.59	68.044192357	converged
naca0012_m080_inviscid	32	10000	0		3.93	80.693608889	converged
naca0012_m200_inviscid	32	2120	0		3	15.537148082000002	converged
naca0012_m015_laminar_re5000	32	10000	0		2.25	96.10867234	converged
naca0012_m080_laminar_re5000	32	10000	0		3.54	113.00992271300001	converged
naca0012_m200_laminar_re5000	32	5503	0		3.7	79.91413564800001	converged
cylinder_m010_laminar_re20	32	10000	0		1.21	56.25327697	converged
cylinder_m010_laminar_re200	32	30000	300		3.28	6685.904943099999	statistically_periodic

Exact command for naca0012_m015_inviscid (source `run_manifest.csv`):

```
mpirun-np32solver / build / cfd_solversolve--casecfd_solver_agentic_benchmark / inputs / cases /
naca0012_m015_inviscid.json--outputsolver/results/naca0012_m015_inviscid--report-levelfull
```

Exact command for naca0012_m080_inviscid (source run_manifest.csv):

```
mpirun-np32solver / build / cfd _ solversolve--casecfd _ solver _ agentic _ benchmark / inputs / cases /
naca0012_m080_inviscid.json--outputsolver/results/naca0012_m080_inviscid--report-levelfull
```

Exact command for naca0012_m200_inviscid (source run_manifest.csv):

```
mpirun-np32solver / build / cfd _ solversolve--casecfd _ solver _ agentic _ benchmark / inputs / cases /
naca0012_m200_inviscid.json--outputsolver/results/naca0012_m200_inviscid--report-levelfull
```

Exact command for naca0012_m015_laminar_re5000 (source run_manifest.csv):

```
mpirun-np32solver / build / cfd _ solversolve--casecfd _ solver _ agentic _ benchmark / inputs /
cases / naca0012 _ m015 _ laminar _ re5000 . json--outputsolver / results / naca0012 _ m015 _ laminar _
re5000--report-levelfull
```

Exact command for naca0012_m080_laminar_re5000 (source run_manifest.csv):

```
mpirun-np32solver / build / cfd _ solversolve--casecfd _ solver _ agentic _ benchmark / inputs /
cases / naca0012 _ m080 _ laminar _ re5000 . json--outputsolver / results / naca0012 _ m080 _ laminar _
re5000--report-levelfull
```

Exact command for naca0012_m200_laminar_re5000 (source run_manifest.csv):

```
mpirun-np32solver / build / cfd _ solversolve--casecfd _ solver _ agentic _ benchmark / inputs /
cases / naca0012 _ m200 _ laminar _ re5000 . json--outputsolver / results / naca0012 _ m200 _ laminar _
re5000--report-levelfull
```

Exact command for cylinder_m010_laminar_re20 (source run_manifest.csv):

```
mpirun-np32solver / build / cfd _ solversolve--casecfd _ solver _ agentic _ benchmark / inputs /
cases / cylinder _ m010 _ laminar _ re20 . json--outputsolver / results / cylinder _ m010 _ laminar _
re20--report-levelfull
```

Exact command for cylinder_m010_laminar_re200 (source run_manifest.csv):

```
mpirun-np32solver / build / cfd _ solversolve--casecfd _ solver _ agentic _ benchmark / inputs /
cases / cylinder _ m010 _ laminar _ re200 . json--outputsolver / results / cylinder _ m010 _ laminar _
re200--report-levelfull
```

8 Results

8.1 Force summary

Steady rows are final recorded samples after residual reduction and a stable terminal-force plateau. Re200 uses a post-transient final-half mean; its lift amplitude is half the corresponding range.

Case	C_D	C_L	C_m	status
naca0012_m015_inviscid	0.14016	-0.0025477	0.00015574	converged
naca0012_m080_inviscid	0.008258	0.0019118	0.00037745	converged
naca0012_m200_inviscid	0.090995	5.9294e-05	-0.000102	converged
naca0012_m015_laminar_re5000	0.43292	-0.0010434	-0.00036509	converged
naca0012_m080_laminar_re5000	0.077839	8.8901e-05	1.8518e-05	converged
naca0012_m200_laminar_re5000	0.040582	-3.087e-06	3.9917e-06	converged
cylinder_m010_laminar_re20	1.9713	0.0042614	-0.00058525	converged
cylinder_m010_laminar_re200	1.1399	-0.00034011	-1.8386e-05	statistically-periodic

The pressure/viscous drag decomposition and transient amplitude are:

Case	pressure drag	viscous drag	C_L amplitude	metric
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naca0012_m015_inviscid	0.14016	0	–	final
naca0012_m080_inviscid	0.008258	0	–	final
naca0012_m200_inviscid	0.090995	0	–	final
naca0012_m015_laminar_re5000	0.0024645	0.43046	–	final
naca0012_m080_laminar_re5000	0.060502	0.017337	–	final
naca0012_m200_laminar_re5000	0.013059	0.027523	–	final
cylinder_m010_laminar_re20	0.41058	1.5607	–	final
cylinder_m010_laminar_re200	0.90421	0.23571	0.33817	mean, final half

9 MPI parallelization and METIS partitioning

Rank 0 builds the dual cell graph with one edge for every interior face and calls METIS k -way partitioning. It then packs and sends each rank only its owned cells, incident faces, boundary tags, and the one-layer off-rank stencil cells needed as ghosts; rank 0 discards the global iterative representation after distribution. Ownership crossings define sorted neighbor lists and exact send/receive global-cell maps. Every halo refresh posts neighbor-scoped `MPI_Irecv/MPI_Isend` operations and waits for completion. Solver iterations never `Allgather` the full mesh or conservative state. Only scalar/vector global diagnostics such as residual norms and integrated forces use `MPI_Allreduce`. Representative production-rank rows below expose owned and ghost cells, boundary faces, neighbor degree, and cell-message counts. Their source is each case’s `partition.diagnostics.csv`; the complete files contain all 32 ranks.

Case	rank	owned	ghost	boundary	neighbors	neighbor ranks	send	receive
naca0012_m200_inviscid	0	665	65	26	3	1;2;3	65	65
naca0012_m200_inviscid	16	662	82	0	7	6;7;17;18;19;22;24	81	82
naca0012_m200_inviscid	31	643	49	24	4	24;25;29;30	49	49
cylinder_m010_laminar_re20	0	318	65	0	4	1;2;3;6	65	65
cylinder_m010_laminar_re20	16	318	73	0	6	17;21;26;27;30;31	73	73
cylinder_m010_laminar_re20	31	323	67	9	6	16;20;21;28;29;30	68	67

9.1 Rank-count validation

The submitted NACA Mach-2 and cylinder Re20 comparisons provide recorded timing, residual, force, owned/ghost, balance, and edge-cut values at two and eight ranks. Increasing ranks reduces owned work per rank but increases the surface-to-volume ratio and halo/message overhead; the available end-to-end timings combine that communication cost with partition and I/O cost and therefore do not isolate it.

Case	ranks	wall s	C_D	C_L	residual metric	owned min/max	ghost mean	balance	edge cut
cylinder_m010_laminar_re20	2	643.87	2.0333	0.003638	280	5092/5093	94.5	1	102
cylinder_m010_laminar_re20	8	163.64	2.0056	0.0020048	280.7	1265/1284	121	1.009	522
naca0012_m200_inviscid	2	204.64	0.091007	7.2975e-05	8.621	10405/10411	119	1	122
naca0012_m200_inviscid	8	52.629	0.091012	6.2526e-05	8.626	2569/2640	96.38	1.015	388

For cylinder_m010_laminar_re20, submitted ranks span 2–8; final-force changes are $\Delta C_D = -0.02769$ and $\Delta C_L = -0.001633$, the recorded residual-metric change is 0.655, and the wall-time ratio is 3.935 (low-rank/high-rank). The load-balance-ratio change is 0.008444; edge cut and ghost counts are recorded rather than used to infer unmeasured communication overhead. For naca0012_m200_inviscid, submitted ranks span 2–8; final-force changes are $\Delta C_D = 5.375e-06$ and $\Delta C_L = -1.045e-05$, the recorded residual-metric change is 0.004949, and the wall-time ratio is 3.888 (low-rank/high-rank). The load-balance-ratio change is 0.01432; edge cut and ghost counts are recorded rather than used to infer unmeasured communication overhead. Exact rank-comparison commands and source CSV paths are retained without abbreviation in `rank.comparison.manifest.csv`.

10 Field, surface, and history results

10.1 naca0012_m015_inviscid

Reported status: **converged**. Final steady evidence gives $C_D = 0.14016$ and $C_L = -0.0025477$ with surface C_p range -2.984–2.079. The small lift at the supplied zero incidence is consistent with upper/lower symmetry. Final cell fields span Mach 0.005075–0.1653 and pressure 30.25–32.79. The near-body views expose the low-Mach pressure field. Slip-normal velocity and zero viscous force are checked directly. The plotted quantity is introduced in Figure 1: Global residual-norm history. The plotted quantity is

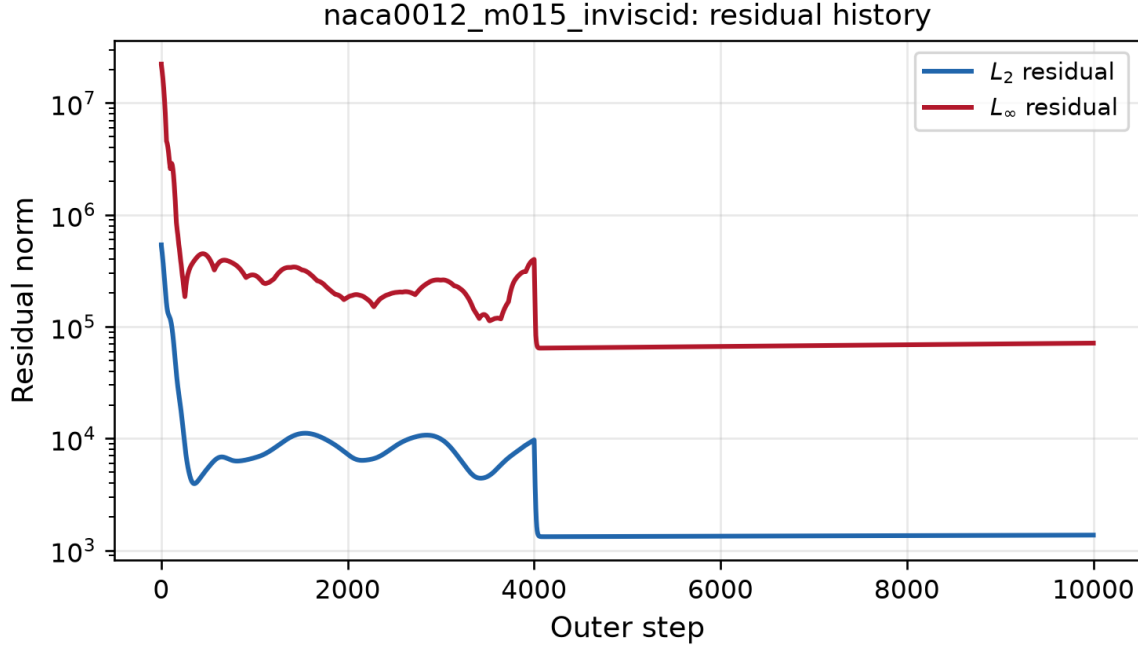


Figure 1: Global residual-norm history.

introduced in Figure 2: Aerodynamic force-coefficient history. The plotted quantity is introduced in Figure 3: Wall pressure coefficient. The plotted quantity is introduced in Figure 4: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 5: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 6: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 7: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

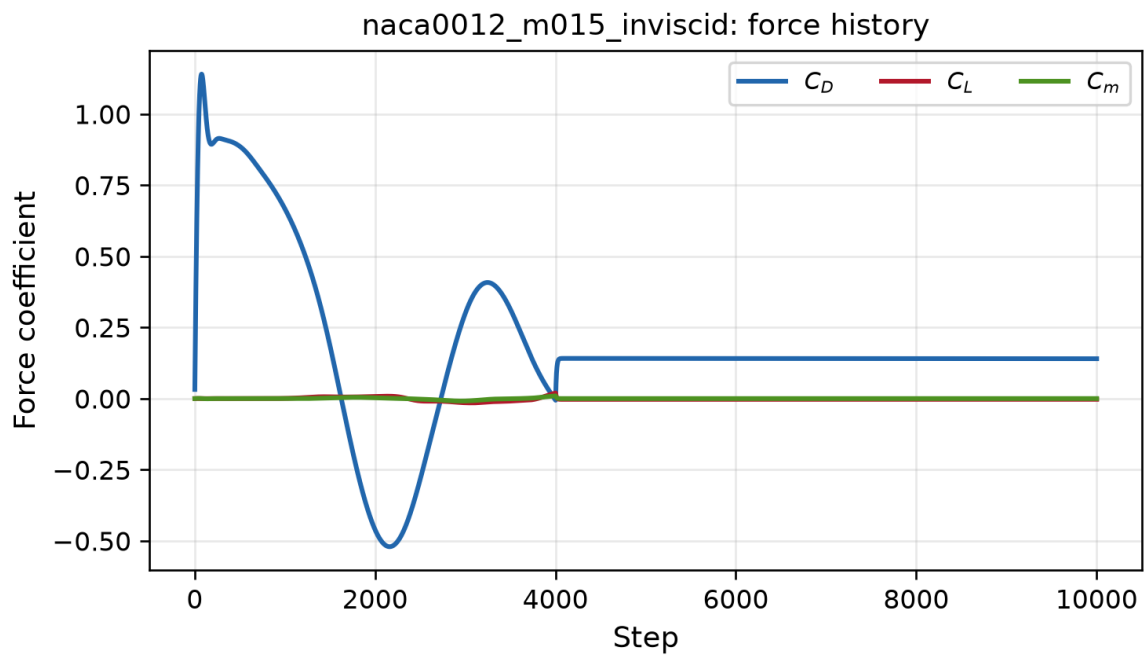


Figure 2: Aerodynamic force-coefficient history.

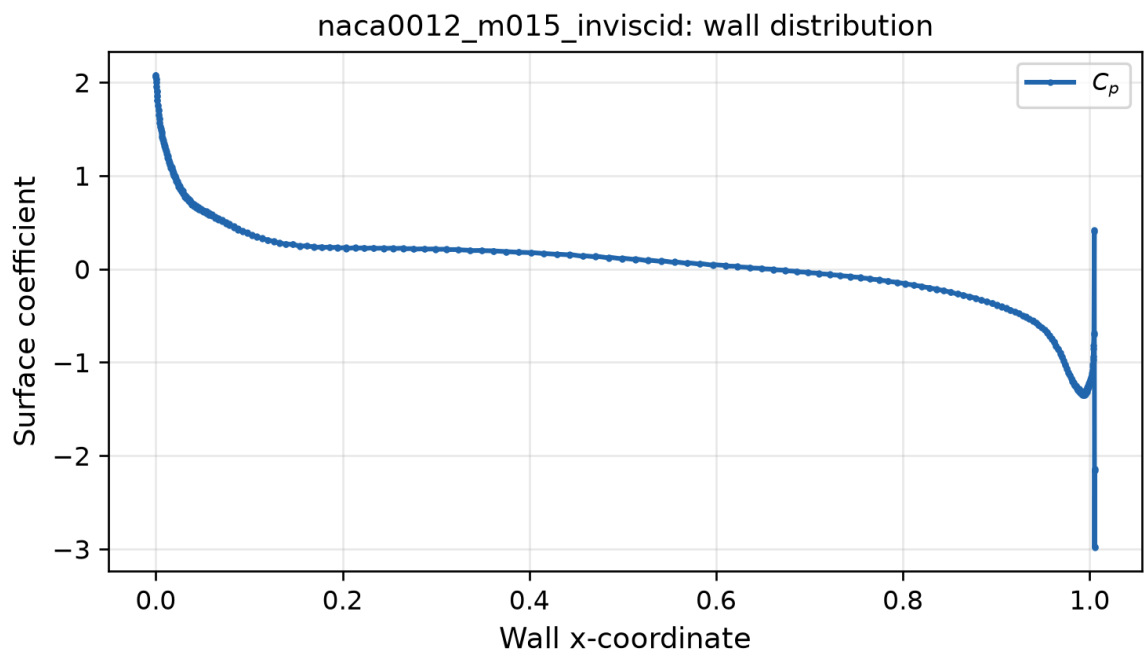


Figure 3: Wall pressure coefficient.

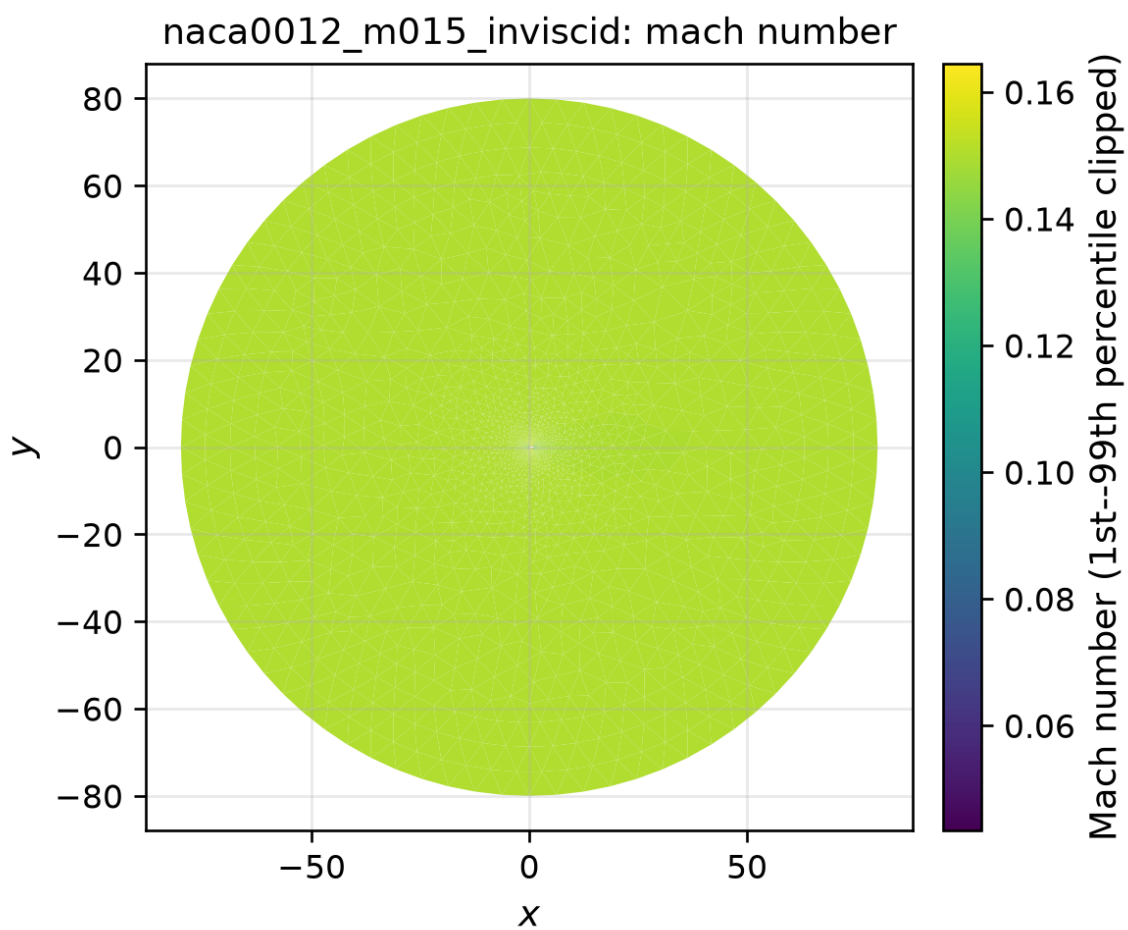


Figure 4: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

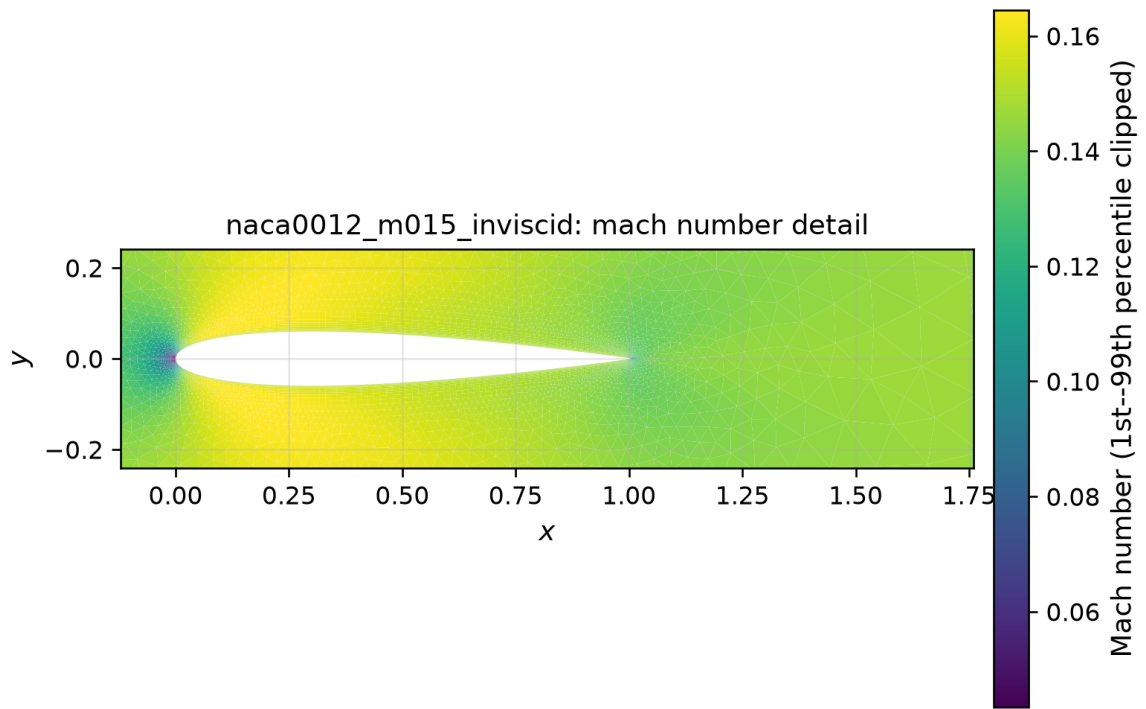


Figure 5: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

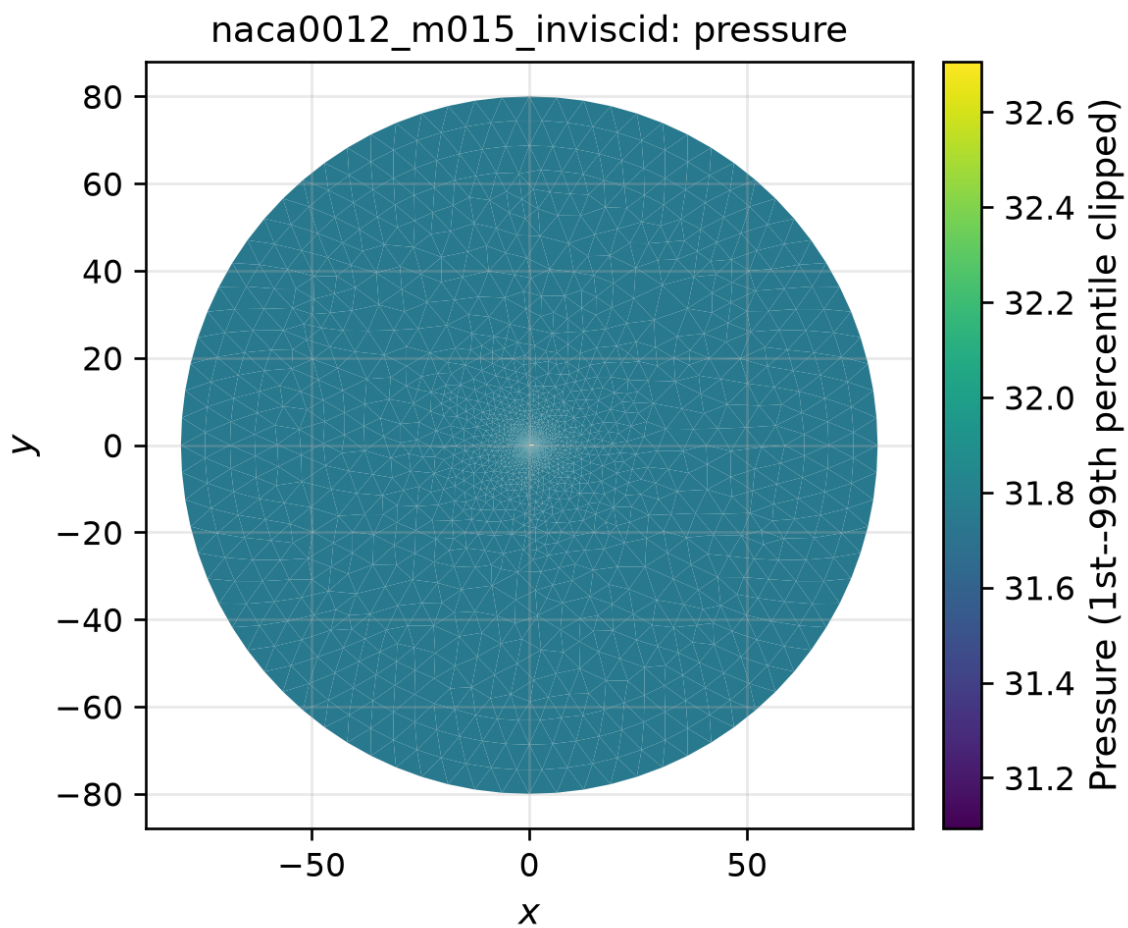


Figure 6: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

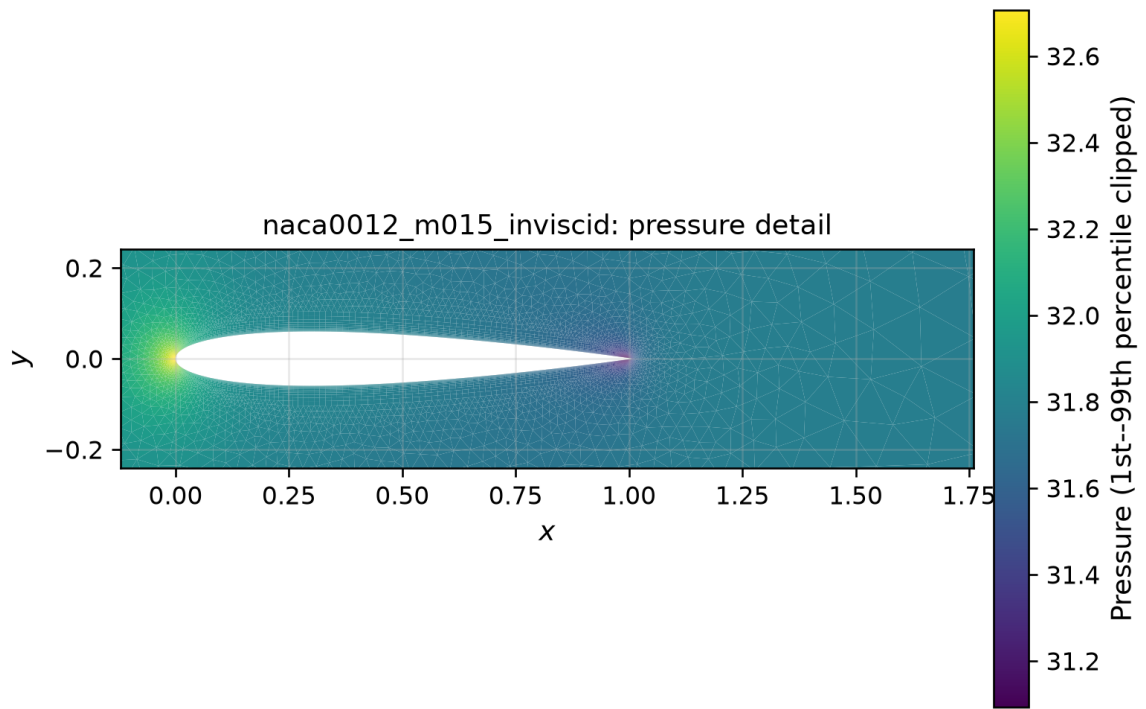


Figure 7: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

10.2 naca0012_m080_inviscid

Reported status: **converged**. Final steady evidence gives $C_D = 0.008258$ and $C_L = 0.0019118$ with surface C_p range -0.939–1.17. The small lift at the supplied zero incidence is consistent with upper/lower symmetry. Final cell fields span Mach 0.02367–1.267 and pressure 0.6466–1.701. The near-body views expose the transonic compressibility and possible localized steep gradients. Slip-normal velocity and zero viscous force are checked directly. The plotted quantity is introduced in Figure 8: Global residual-norm history. The plotted quantity is introduced in Figure 9: Aerodynamic force-coefficient history. The

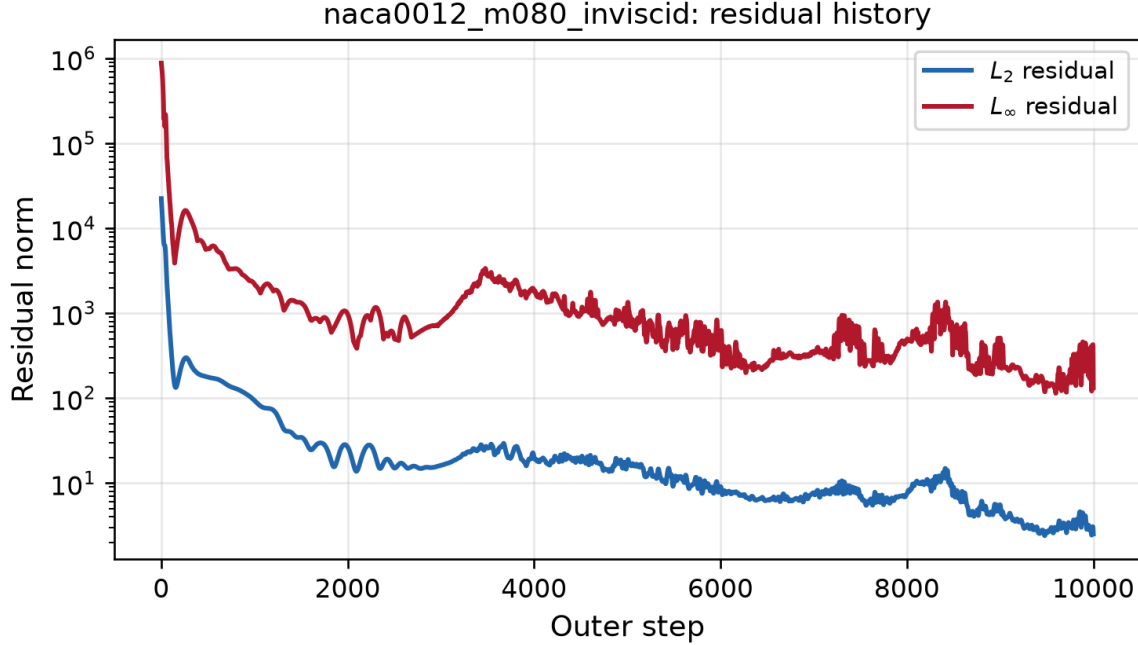


Figure 8: Global residual-norm history.

plotted quantity is introduced in Figure 10: Wall pressure coefficient. The plotted quantity is introduced in Figure 11: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 12: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 13: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 14: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

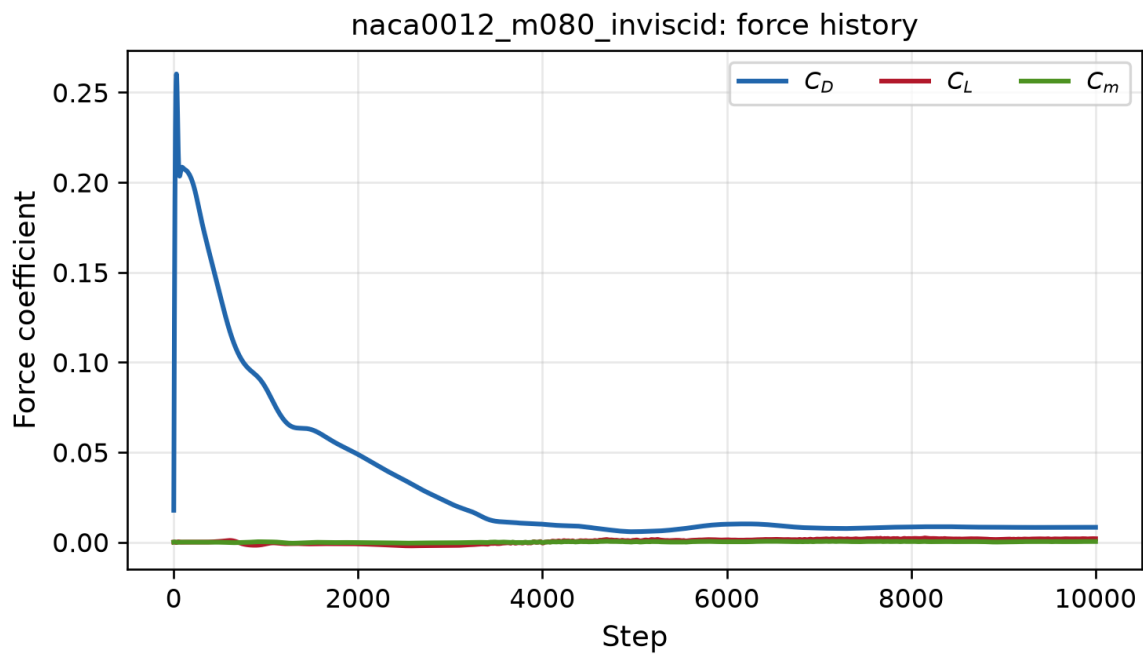


Figure 9: Aerodynamic force-coefficient history.

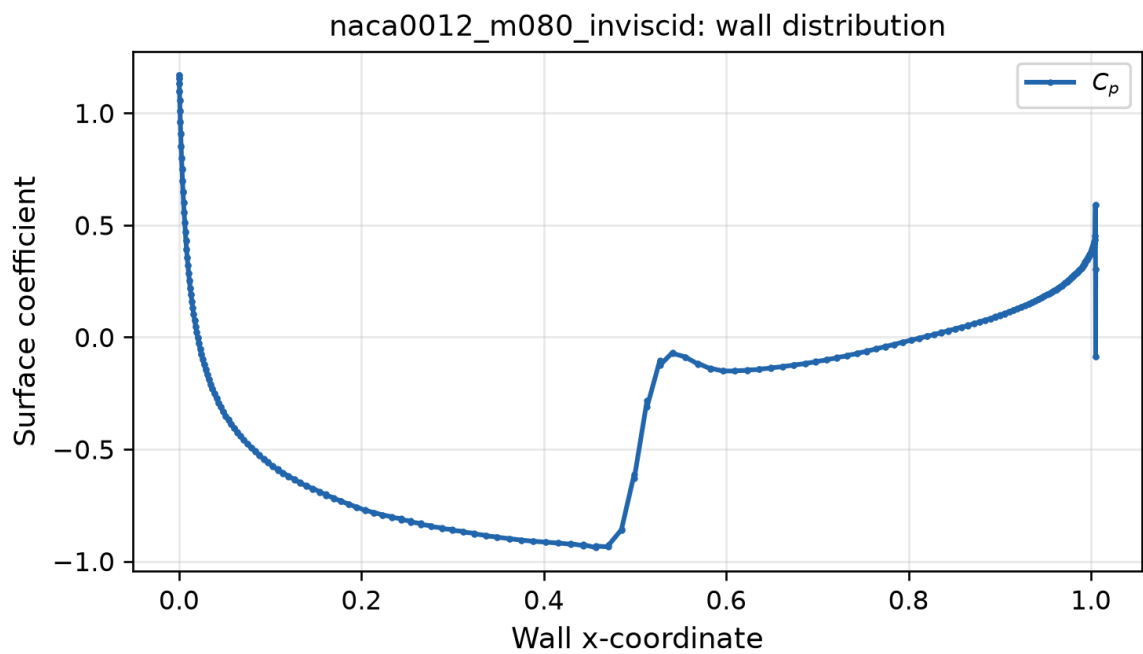


Figure 10: Wall pressure coefficient.

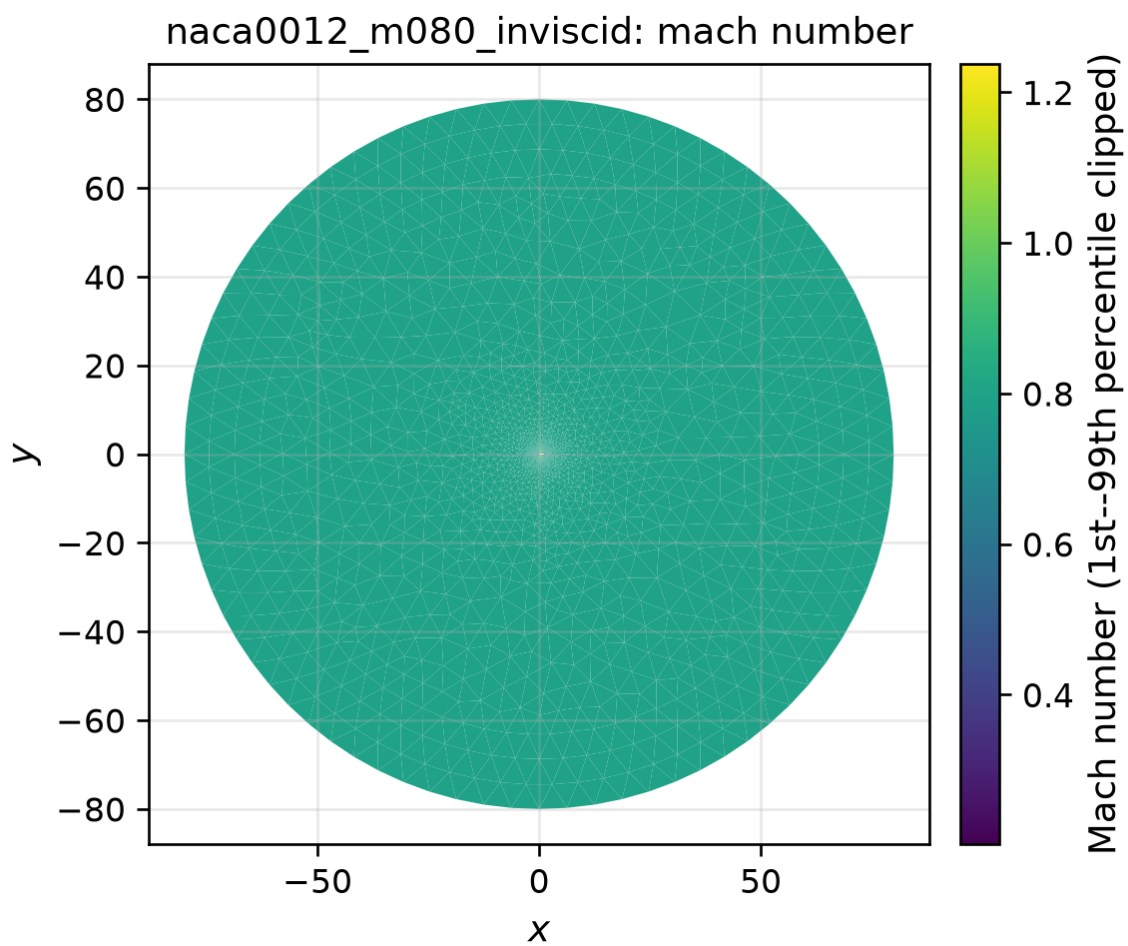


Figure 11: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

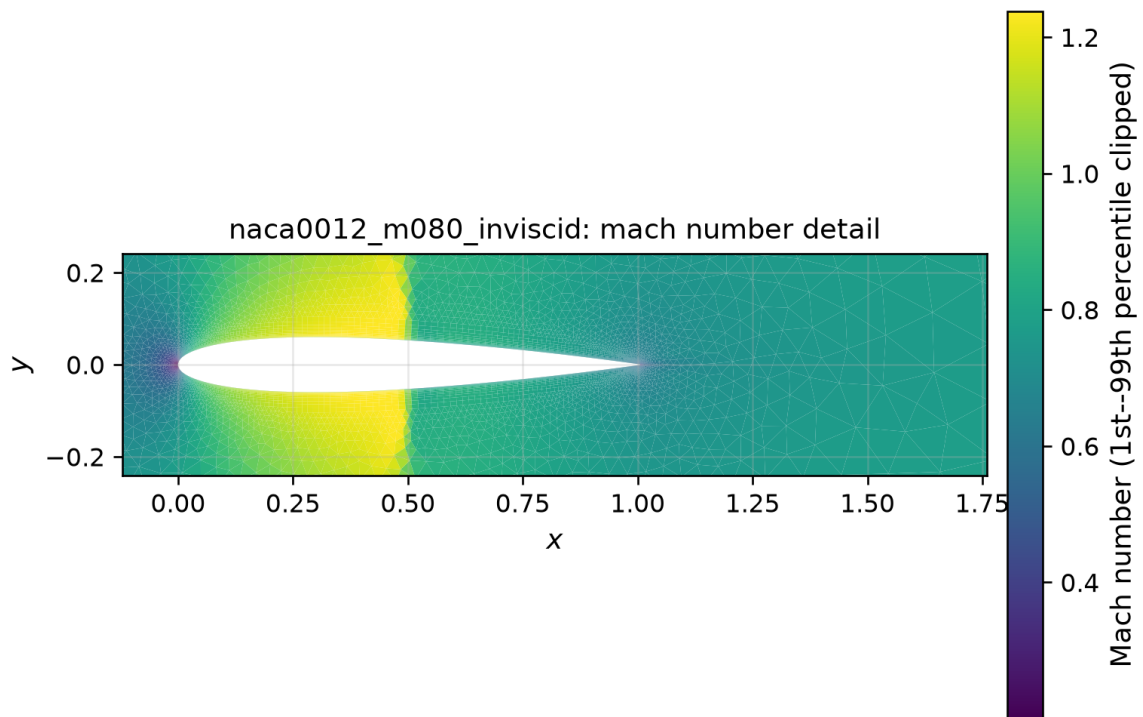


Figure 12: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

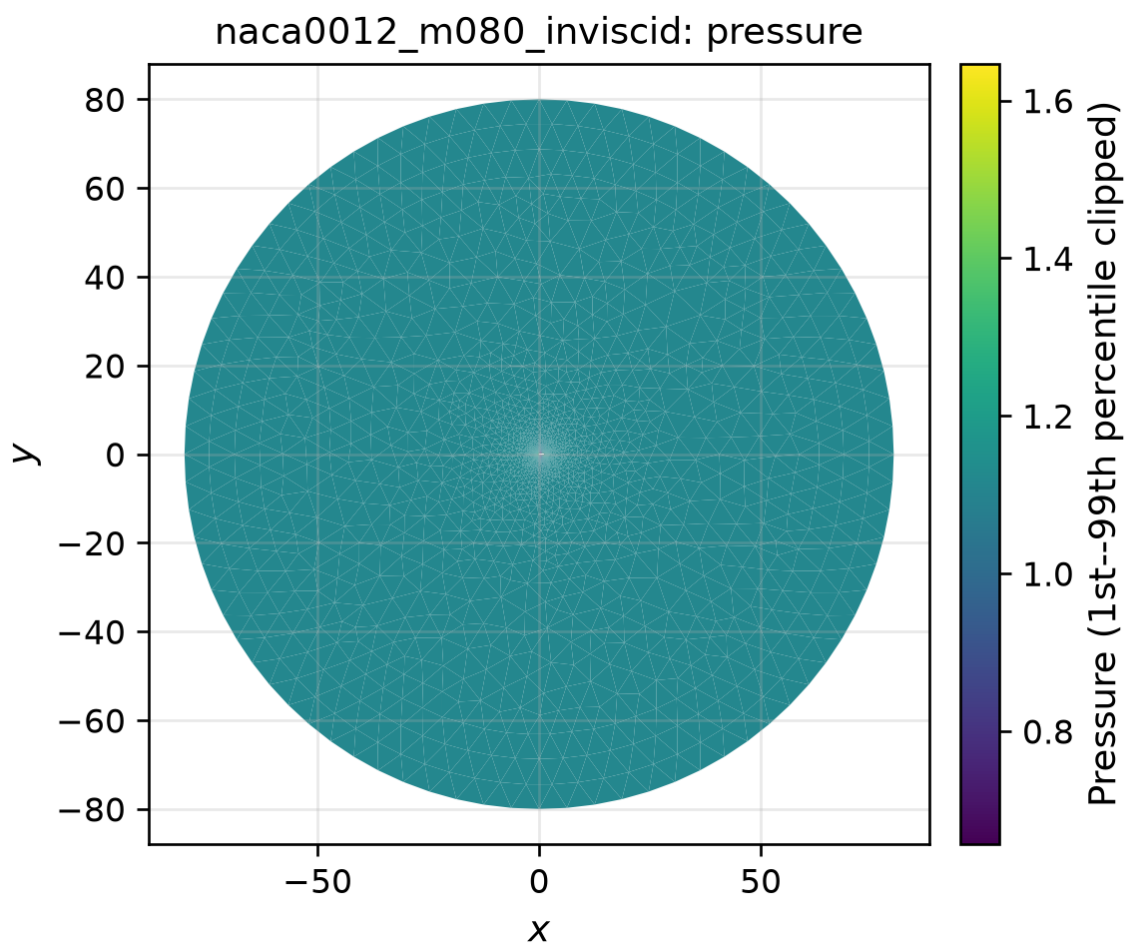


Figure 13: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

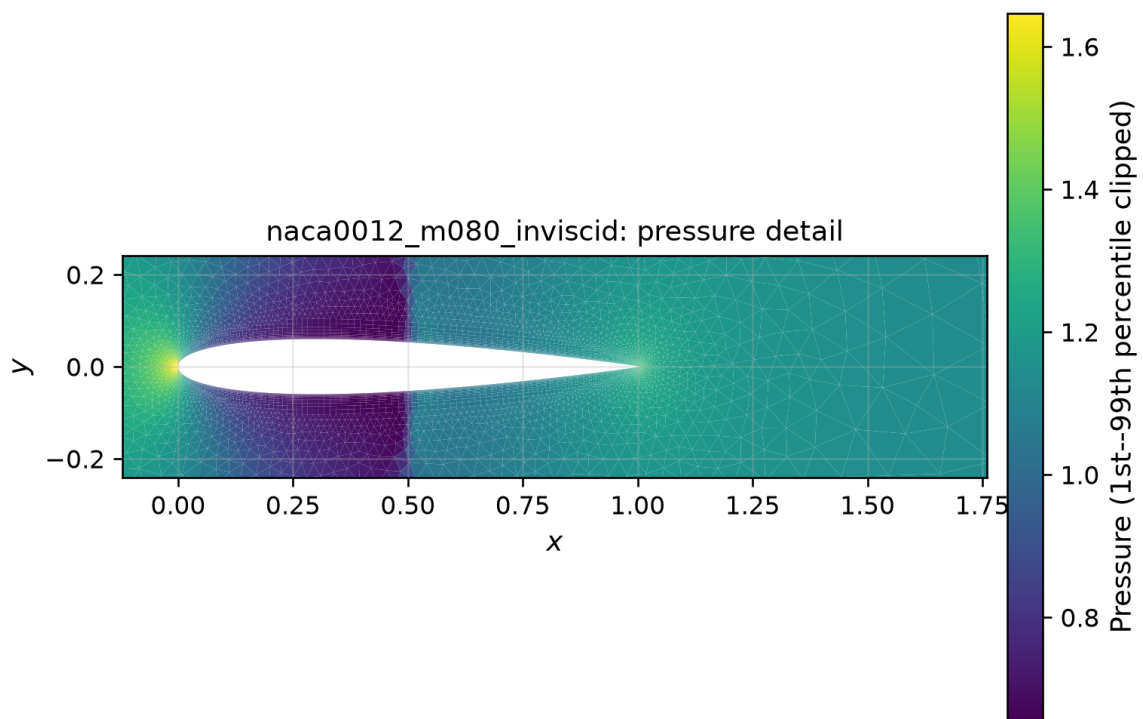


Figure 14: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

10.3 naca0012_m200_inviscid

Reported status: **converged**. Final steady evidence gives $C_D = 0.090995$ and $C_L = 5.9294e - 05$ with surface C_p range -0.3501 – 1.724 . The small lift at the supplied zero incidence is consistent with upper/lower symmetry. Final cell fields span Mach 0.02819 – 2.265 and pressure 0.003546 – 1.041 . The near-body views expose the supersonic compression/expansion structure, with shocks broadened by Rusanov dissipation. Slip-normal velocity and zero viscous force are checked directly. The plotted quantity is introduced in Figure 15: Global residual-norm history. The plotted quantity is introduced in Figure 16: Aerodynamic

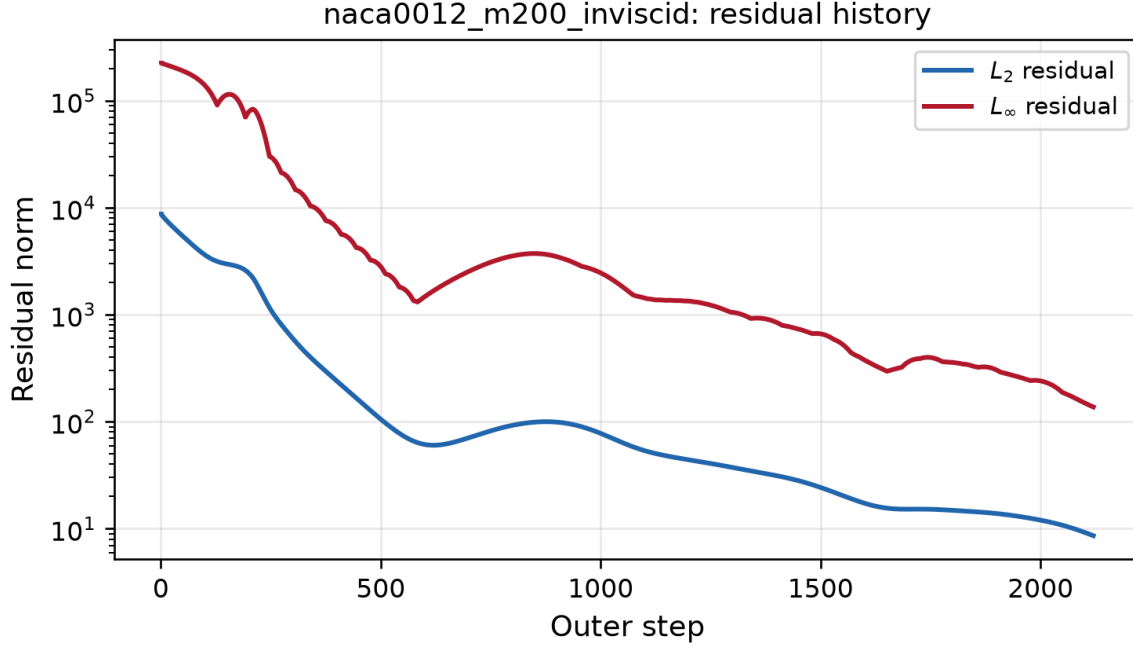


Figure 15: Global residual-norm history.

force-coefficient history. The plotted quantity is introduced in Figure 17: Wall pressure coefficient. The plotted quantity is introduced in Figure 18: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 19: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 20: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 21: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

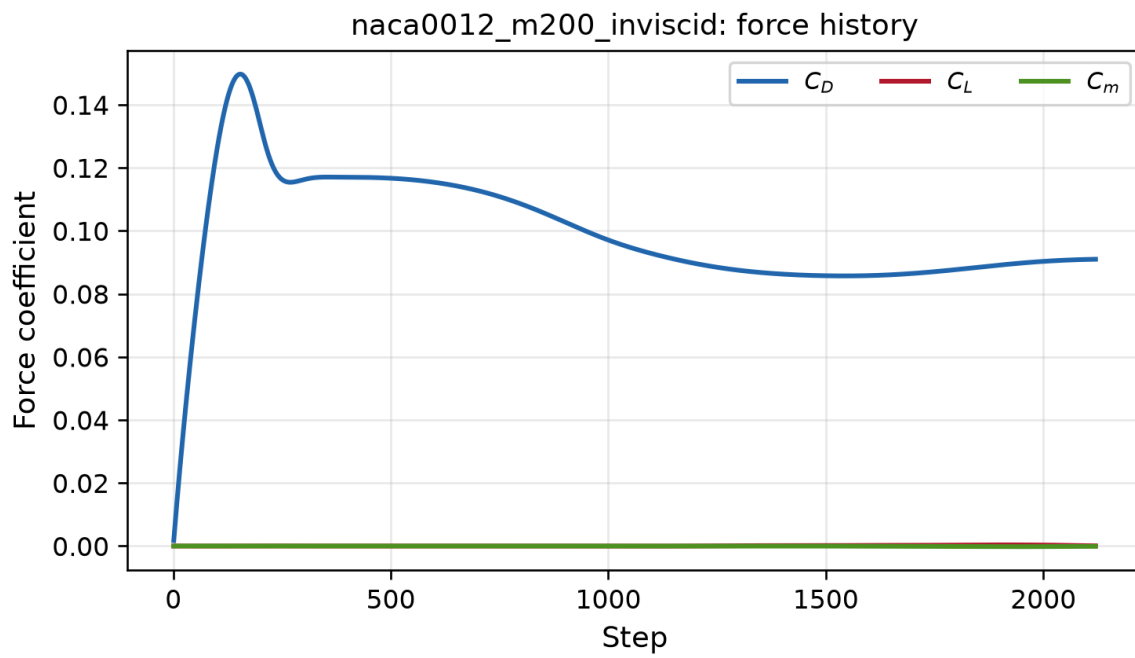


Figure 16: Aerodynamic force-coefficient history.

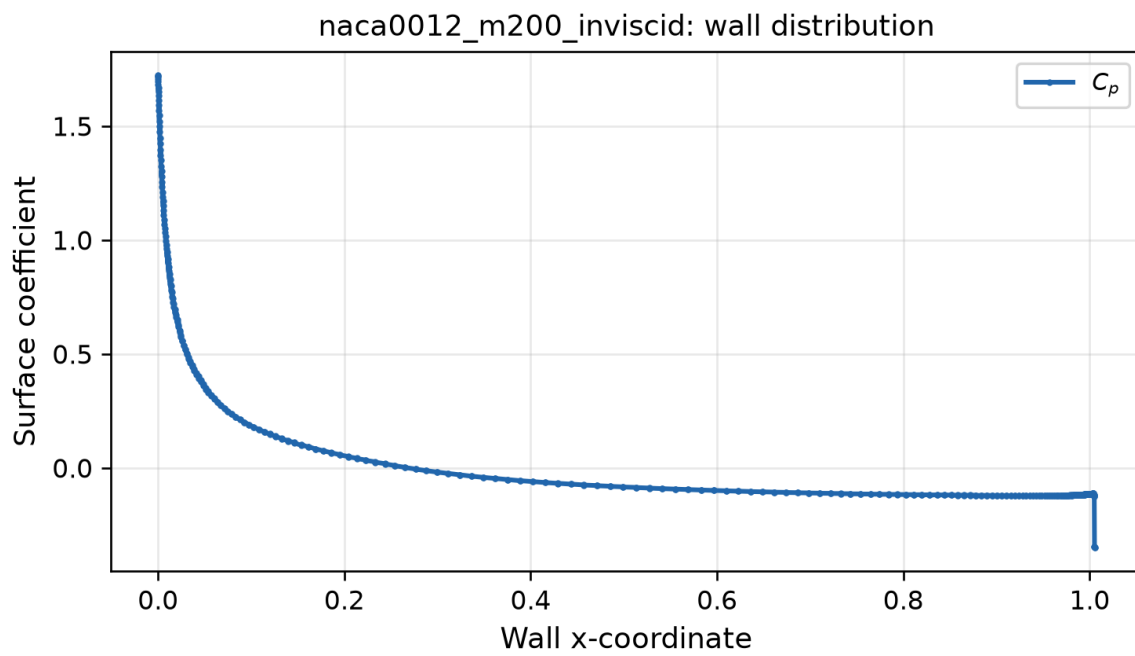


Figure 17: Wall pressure coefficient.

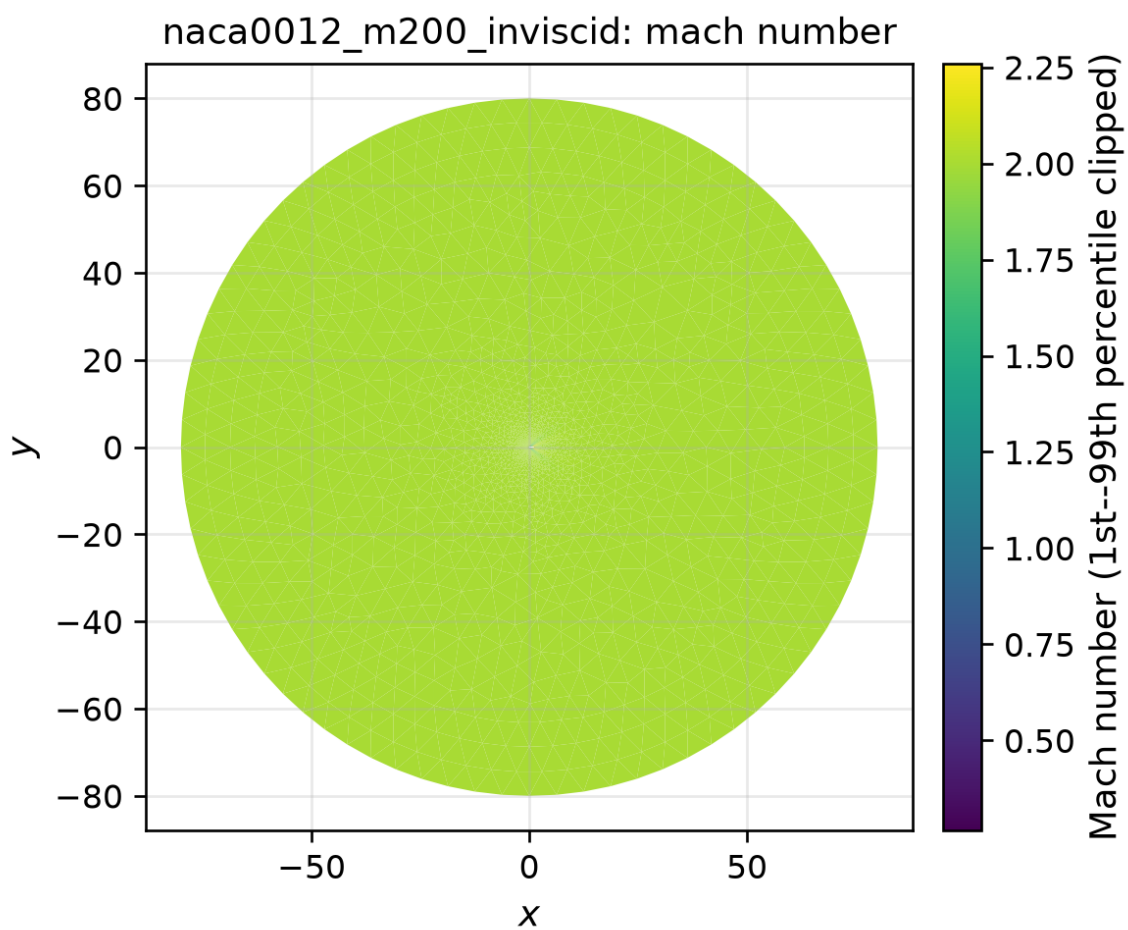


Figure 18: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

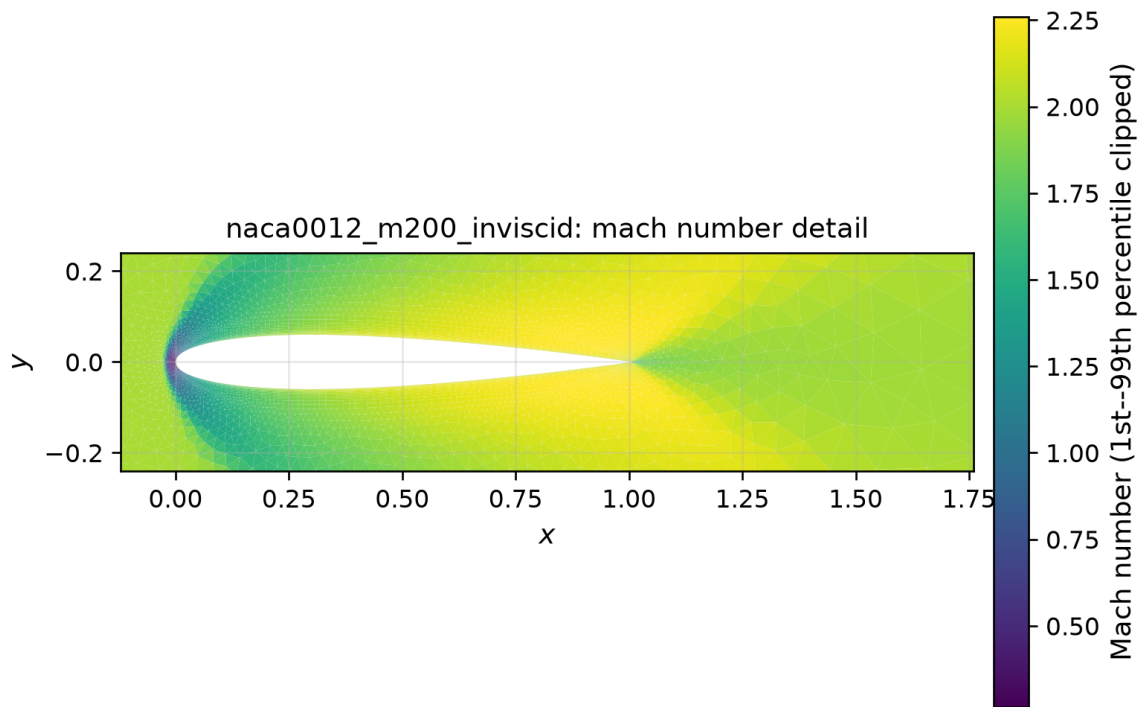


Figure 19: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

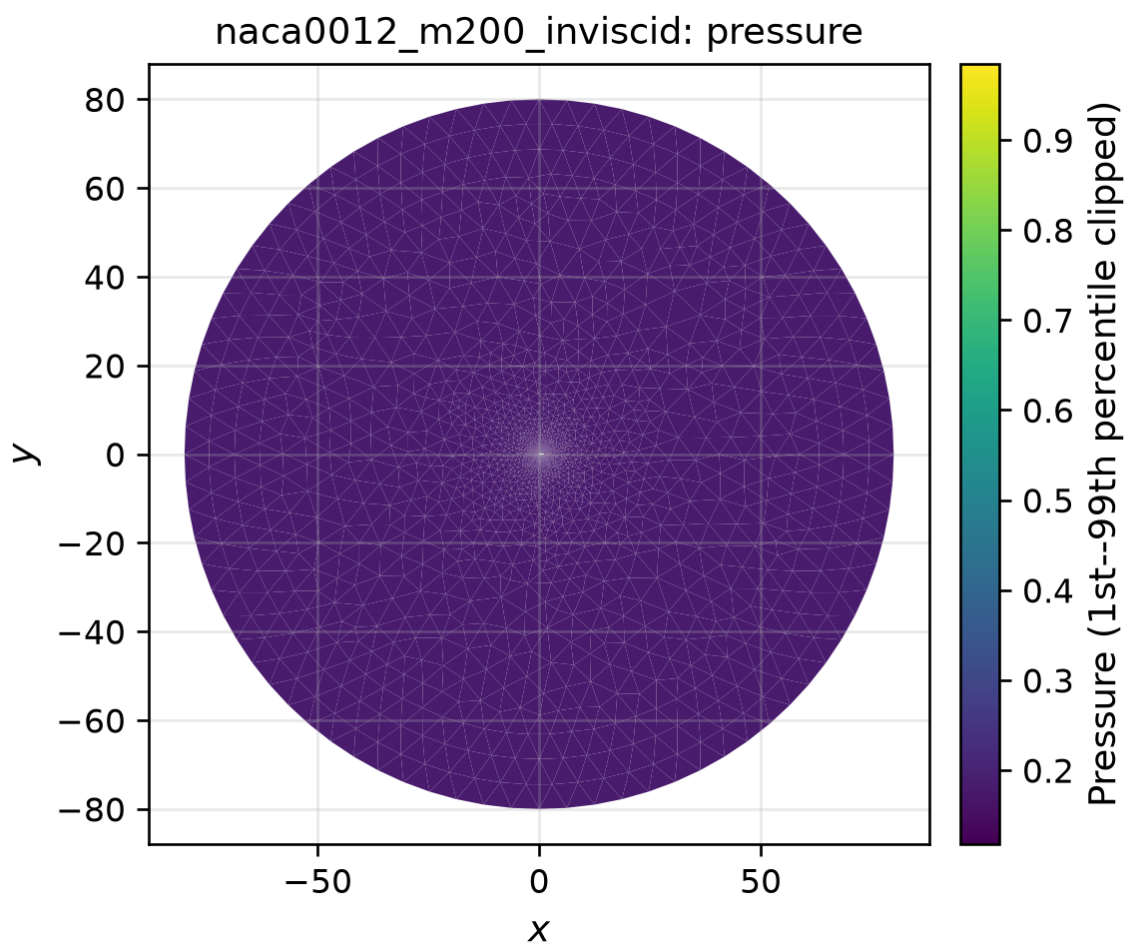


Figure 20: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

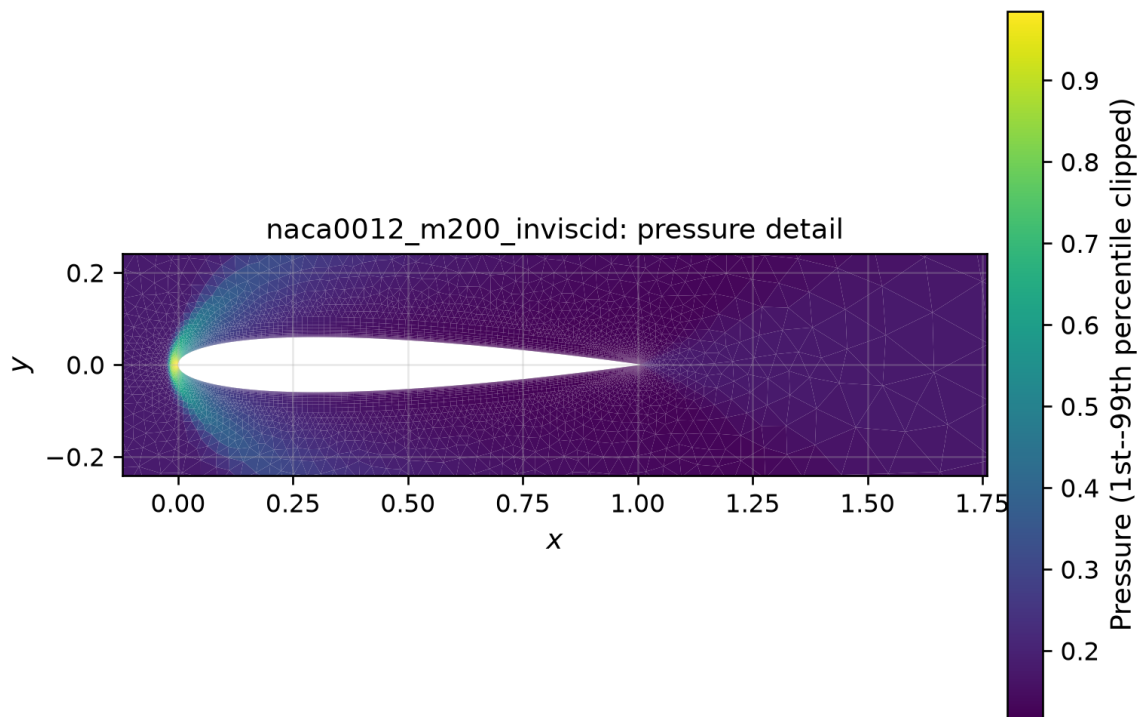


Figure 21: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

10.4 naca0012_m015_laminar_re5000

Reported status: **converged**. Final steady evidence gives $C_D = 0.43292$ and $C_L = -0.0010434$ with surface C_p range -0.9652 – -0.9606 . The small lift at the supplied zero incidence is consistent with upper/lower symmetry. Final cell fields span Mach $1.646\text{e-}05$ – 0.1728 and pressure 31.26 – 32.27 . The near-body views expose the low-Mach pressure field. No-slip wall speed and nonzero skin friction are checked directly. The plotted quantity is introduced in Figure 22: Global residual-norm history. The plotted quantity is

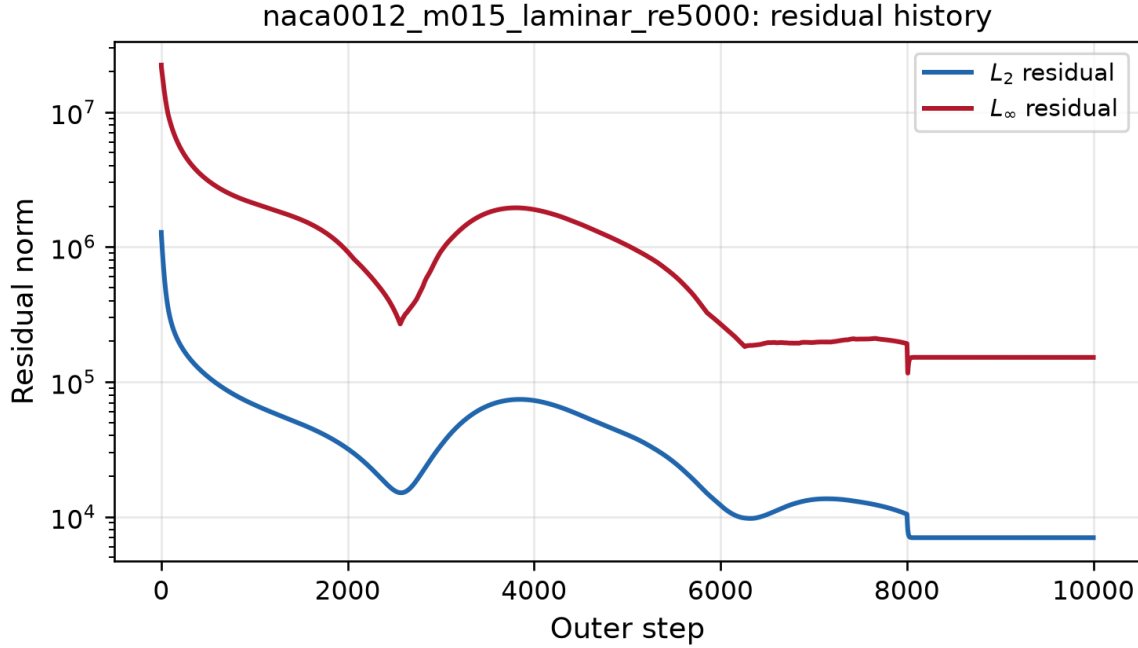


Figure 22: Global residual-norm history.

introduced in Figure 23: Aerodynamic force-coefficient history. The plotted quantity is introduced in Figure 24: Wall pressure coefficient and skin-friction coefficient. The plotted quantity is introduced in Figure 25: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 26: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 27: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 28: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

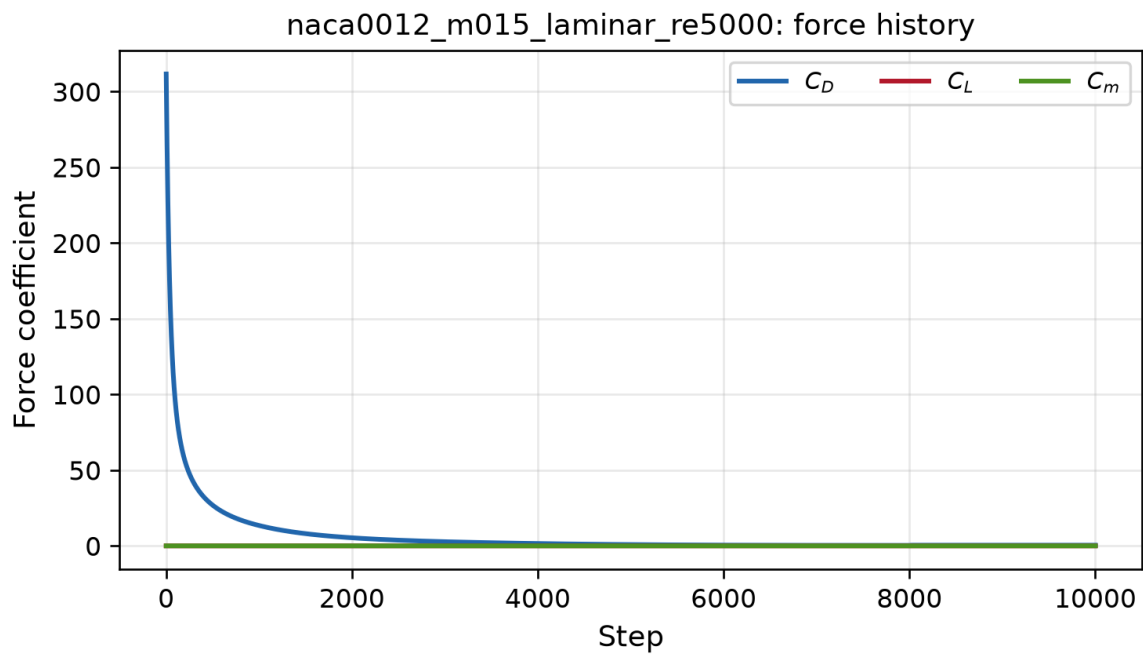


Figure 23: Aerodynamic force-coefficient history.

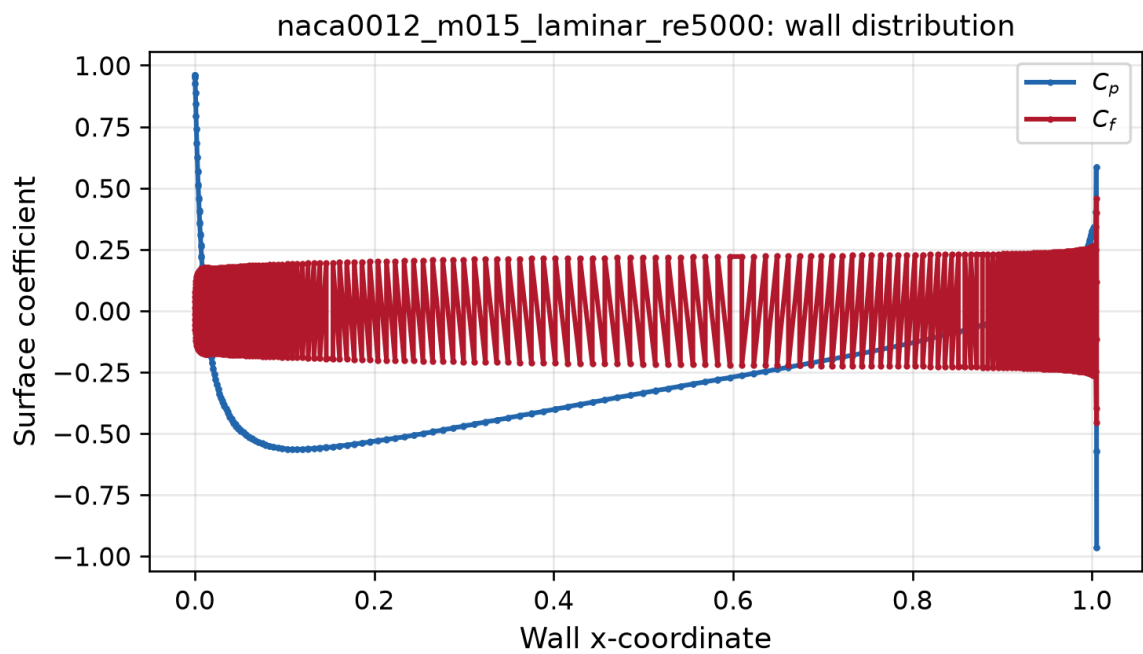


Figure 24: Wall pressure coefficient and skin-friction coefficient.

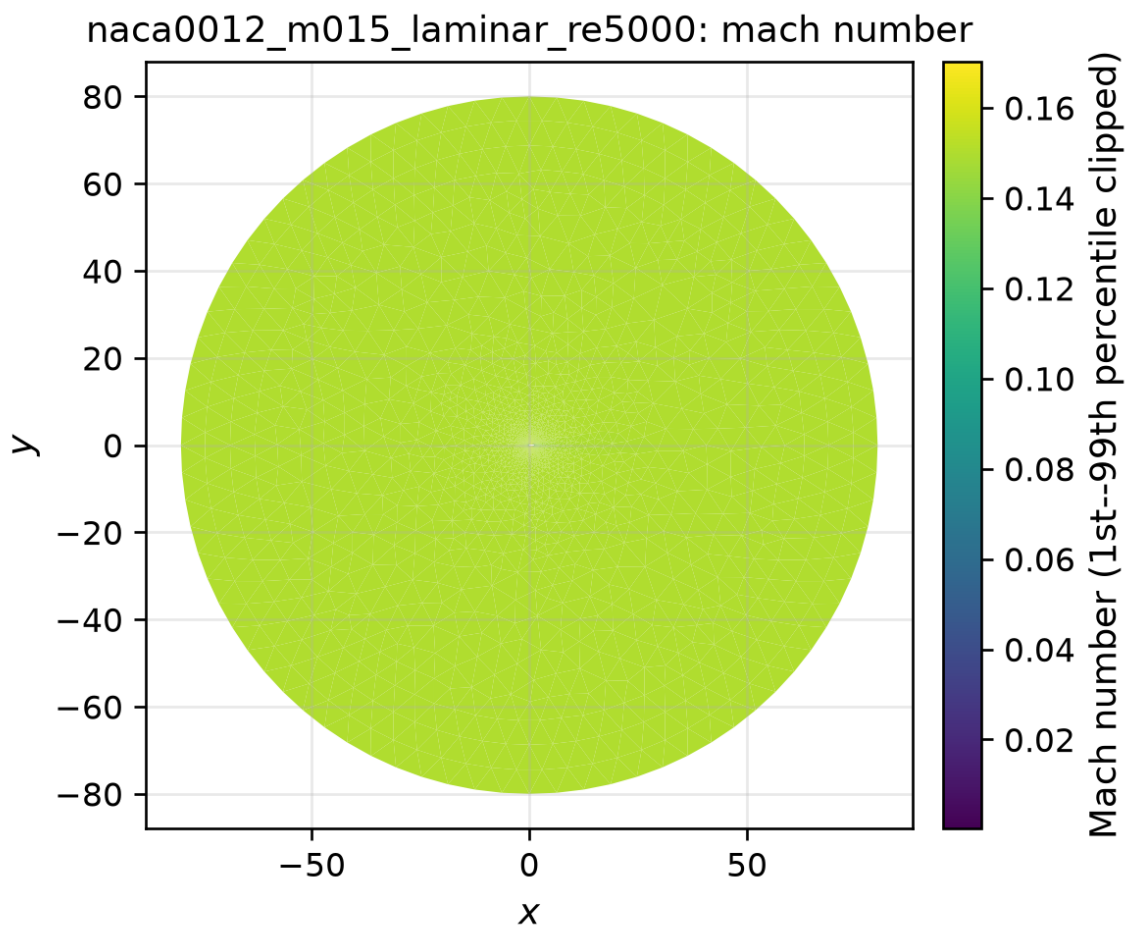


Figure 25: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

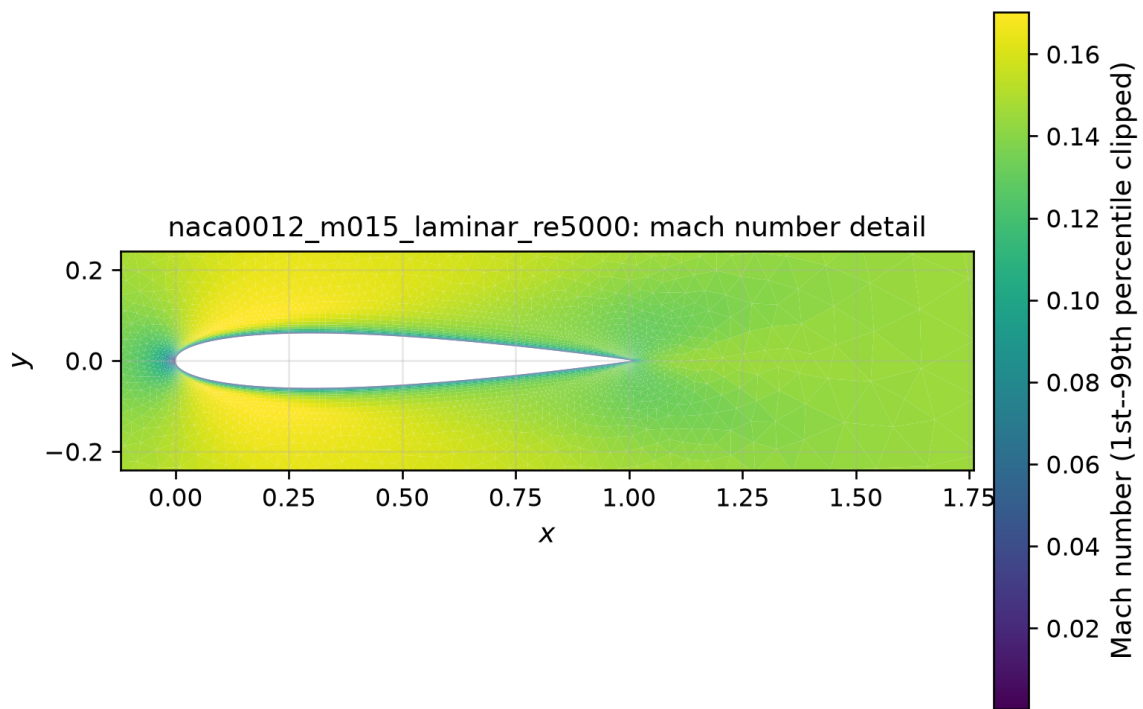


Figure 26: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

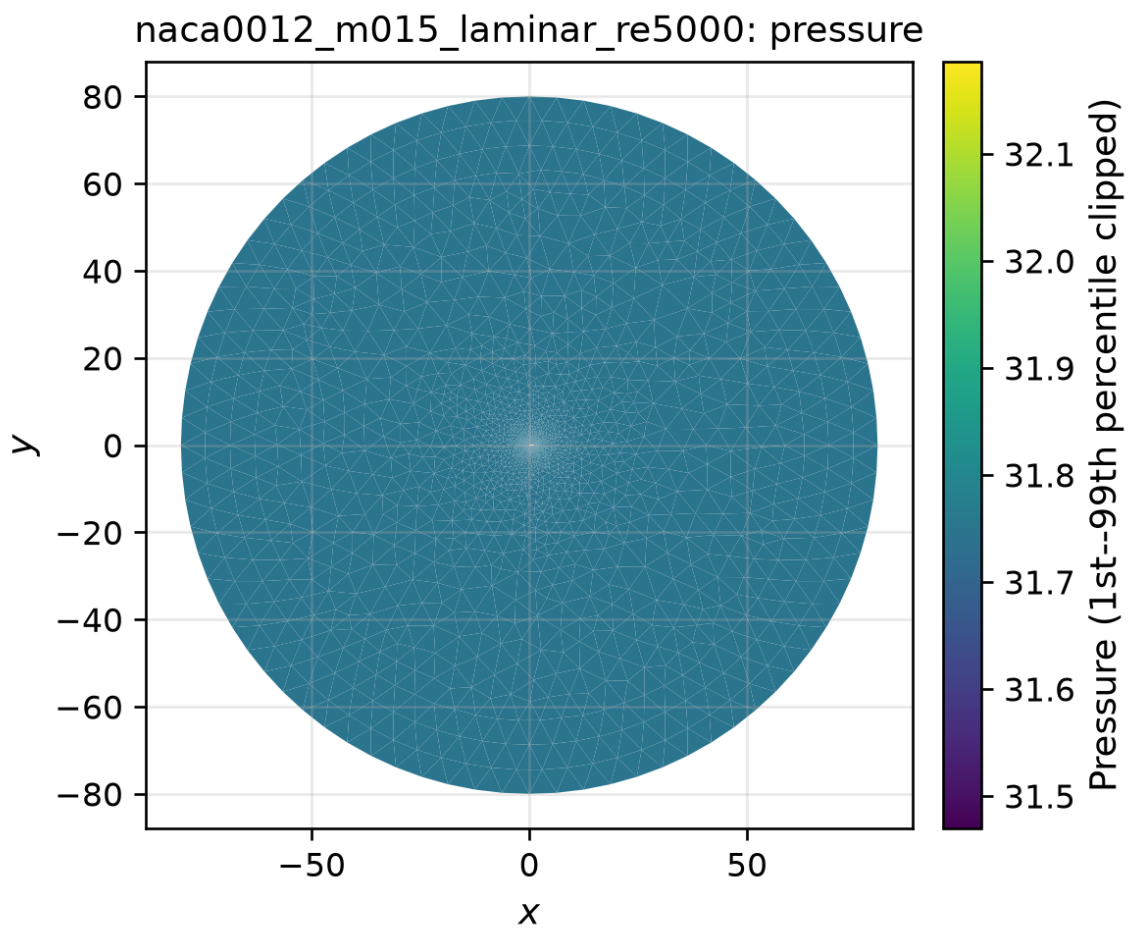


Figure 27: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

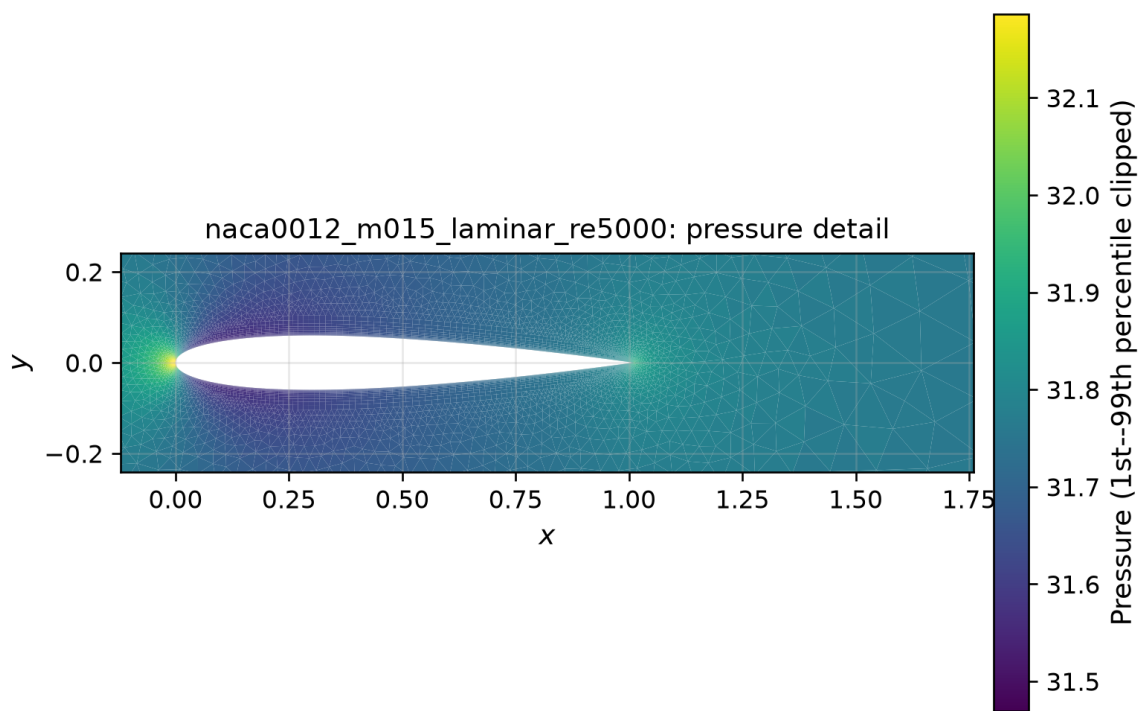


Figure 28: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

10.5 naca0012_m080_laminar_re5000

Reported status: **converged**. Final steady evidence gives $C_D = 0.077839$ and $C_L = 8.8901e - 05$ with surface C_p range -0.2223 – 1.082 . The small lift at the supplied zero incidence is consistent with upper/lower symmetry. Final cell fields span Mach 0.0003484 – 0.9814 and pressure 0.9053 – 1.706 . The near-body views expose the transonic compressibility and possible localized steep gradients. No-slip wall speed and nonzero skin friction are checked directly. The plotted quantity is introduced in Figure 29: Global residual-norm history. The plotted quantity is introduced in Figure 30: Aerodynamic force-coefficient history. The

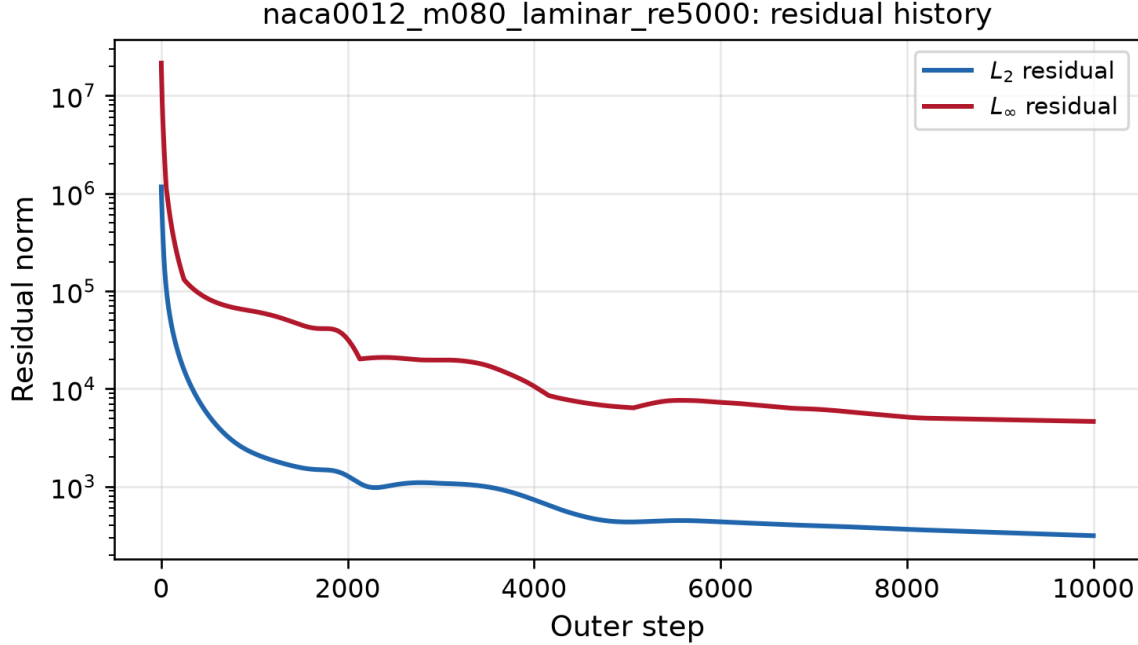


Figure 29: Global residual-norm history.

plotted quantity is introduced in Figure 31: Wall pressure coefficient and skin-friction coefficient. The plotted quantity is introduced in Figure 32: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 33: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 34: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 35: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

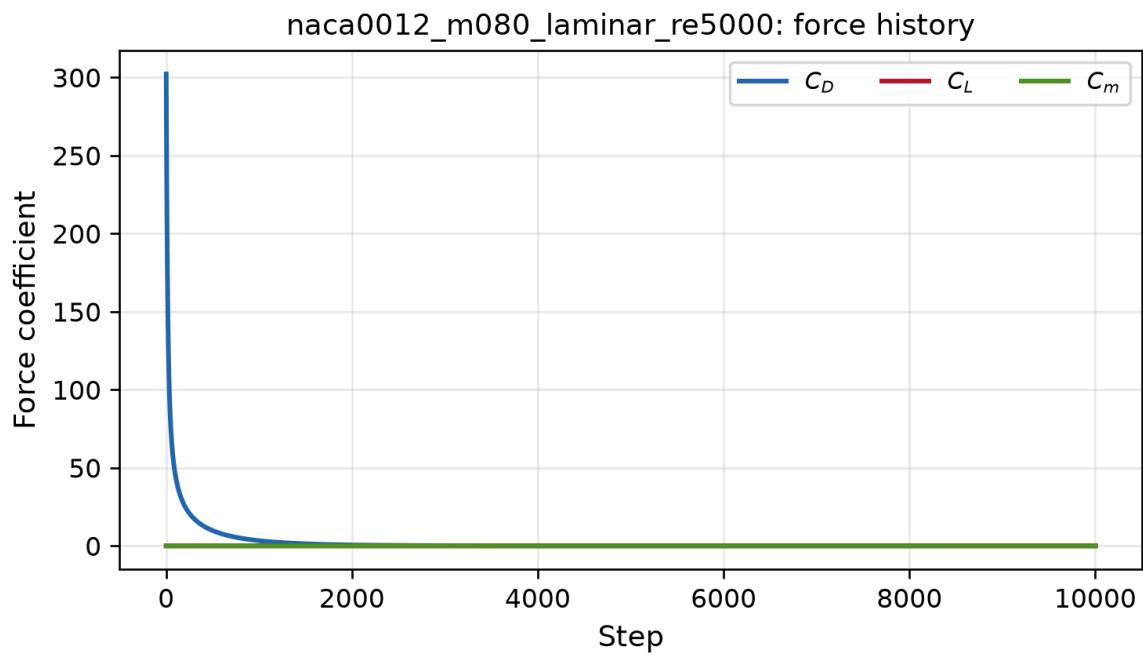


Figure 30: Aerodynamic force-coefficient history.

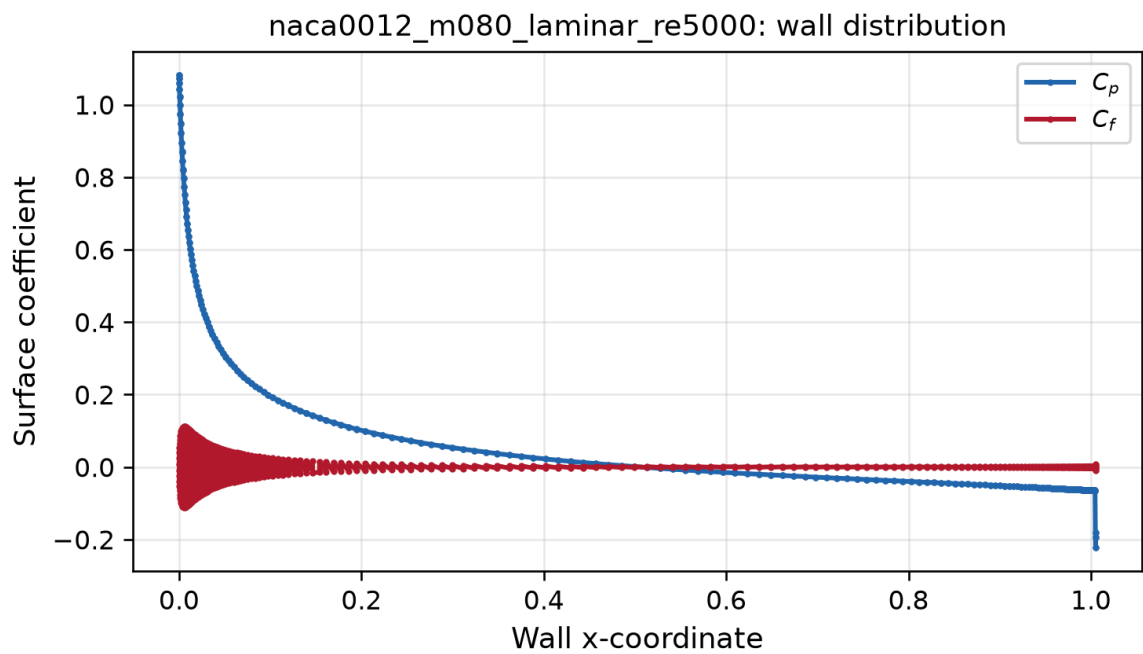


Figure 31: Wall pressure coefficient and skin-friction coefficient.

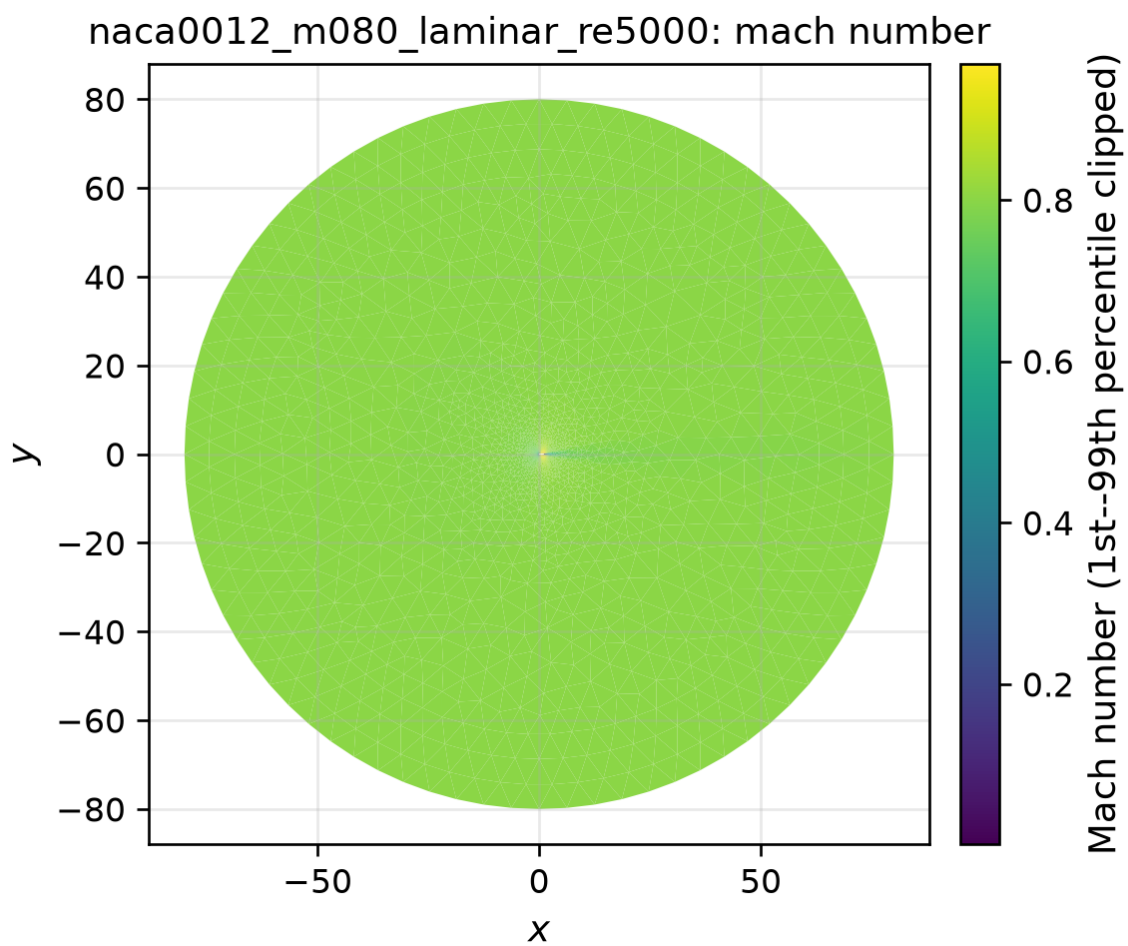


Figure 32: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

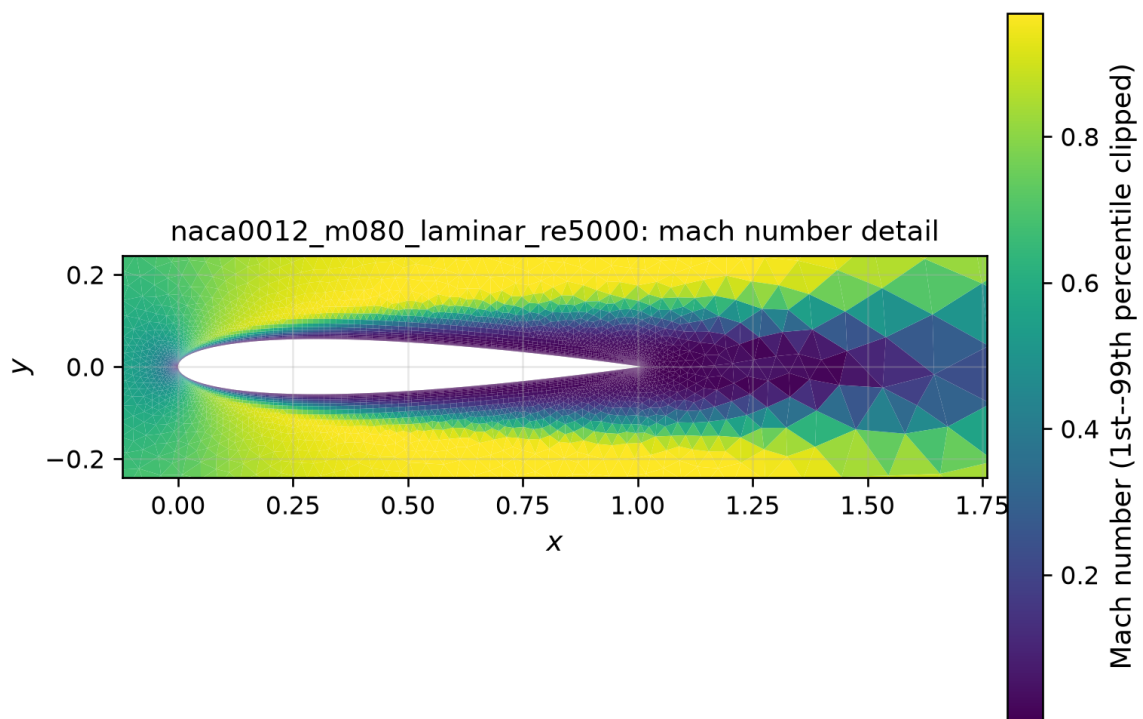


Figure 33: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st-99th percentiles.

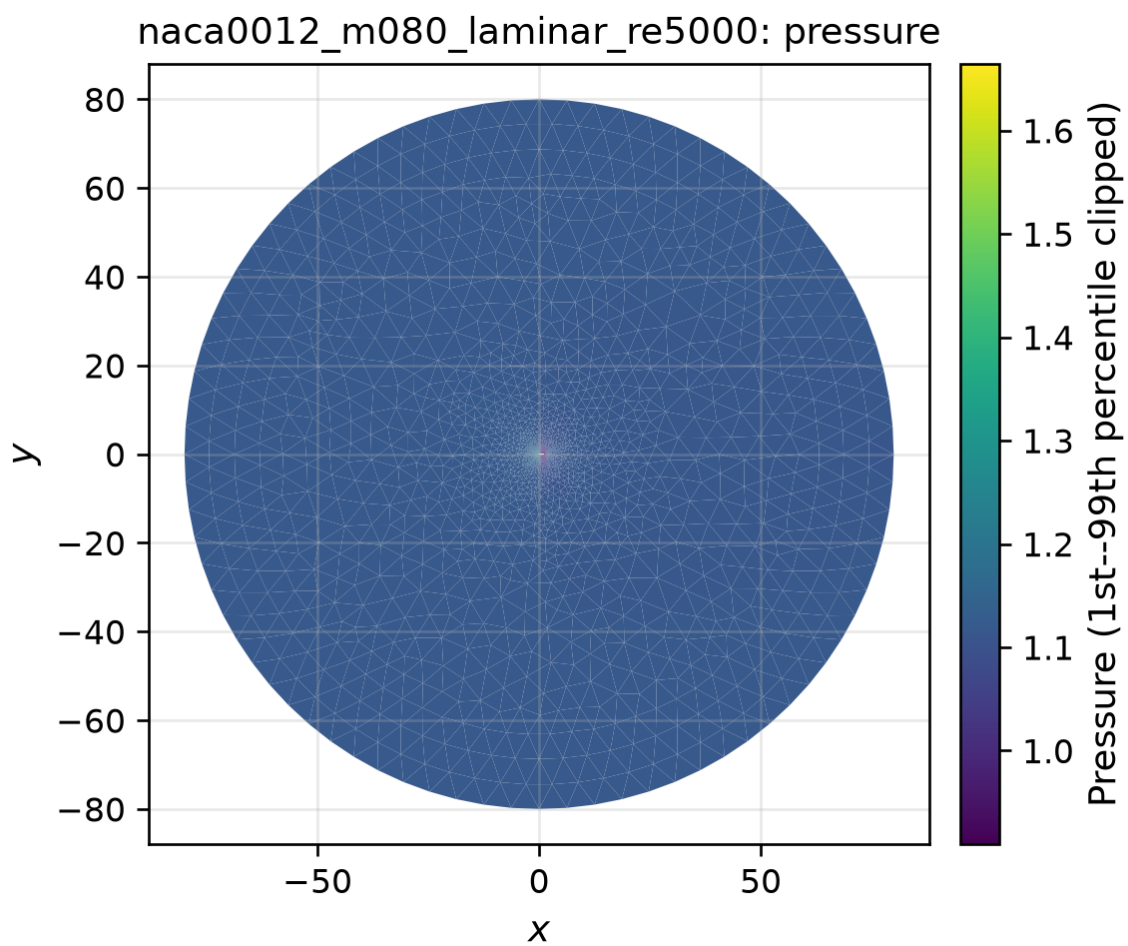


Figure 34: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

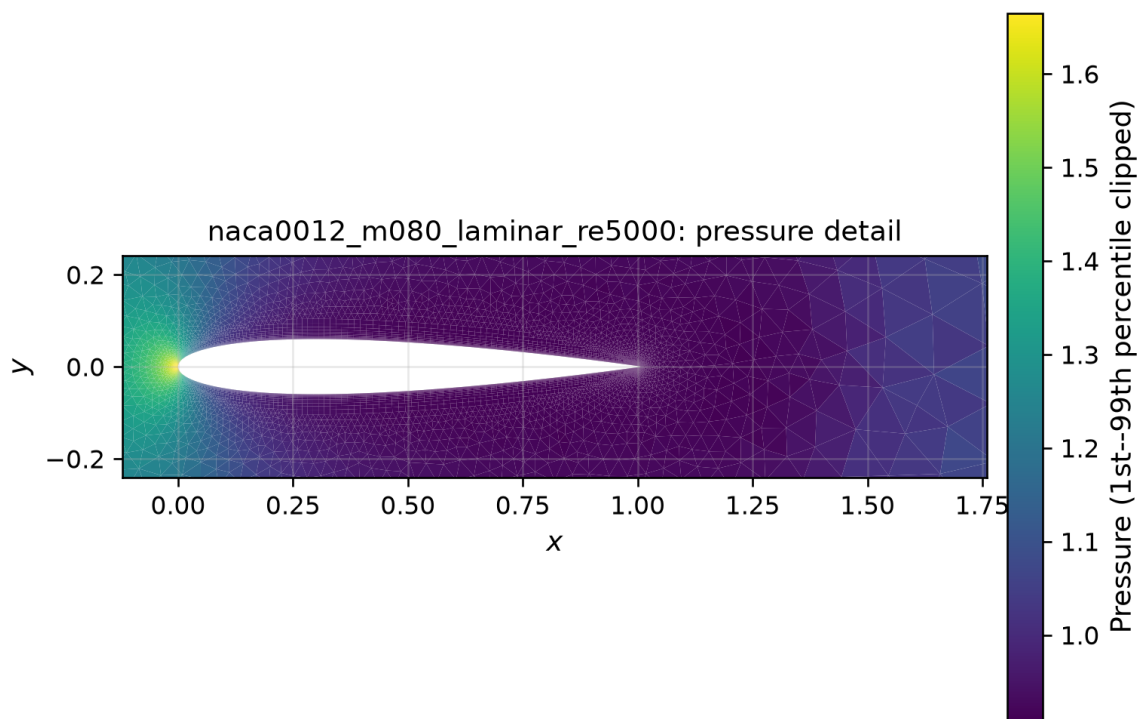


Figure 35: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

10.6 naca0012_m200_laminar_re5000

Reported status: **converged**. Final steady evidence gives $C_D = 0.040582$ and $C_L = -3.087e - 06$ with surface C_p range 0.2504–0.6. The small lift at the supplied zero incidence is consistent with upper/lower symmetry. Final cell fields span Mach 0.0003823–2.05 and pressure 0.09687–1.02. The near-body views expose the supersonic compression/expansion structure, with shocks broadened by Rusanov dissipation. No-slip wall speed and nonzero skin friction are checked directly. The plotted quantity is introduced in Figure 36: Global residual-norm history. The plotted quantity is introduced in Figure 37: Aerodynamic

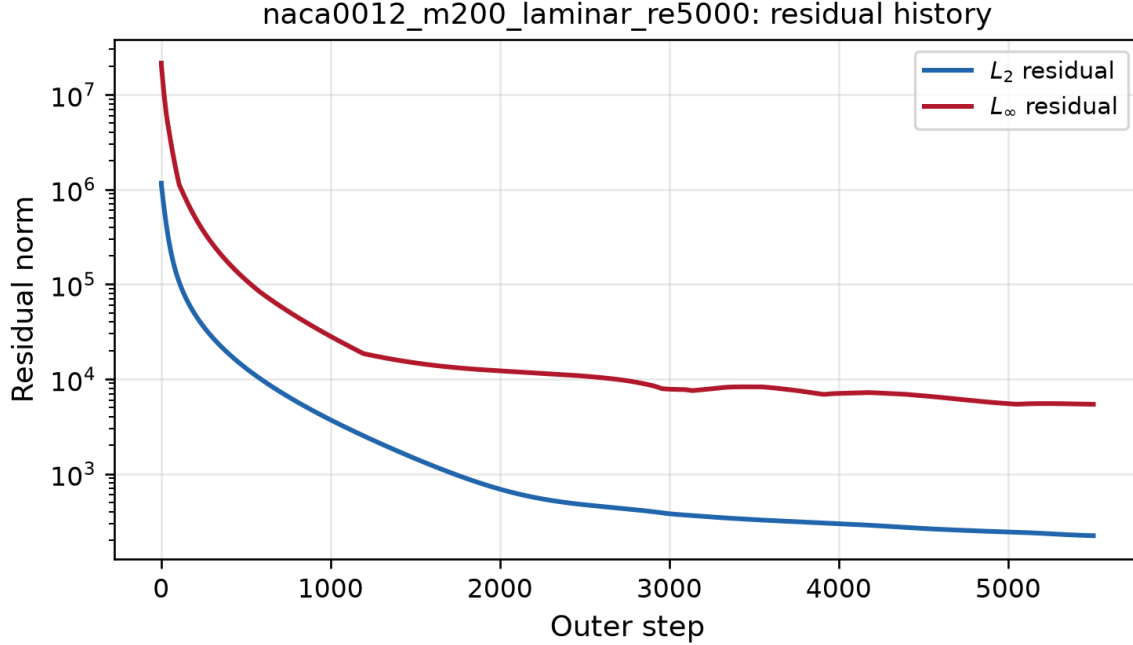


Figure 36: Global residual-norm history.

force-coefficient history. The plotted quantity is introduced in Figure 38: Wall pressure coefficient and skin-friction coefficient. The plotted quantity is introduced in Figure 39: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 40: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 41: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 42: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

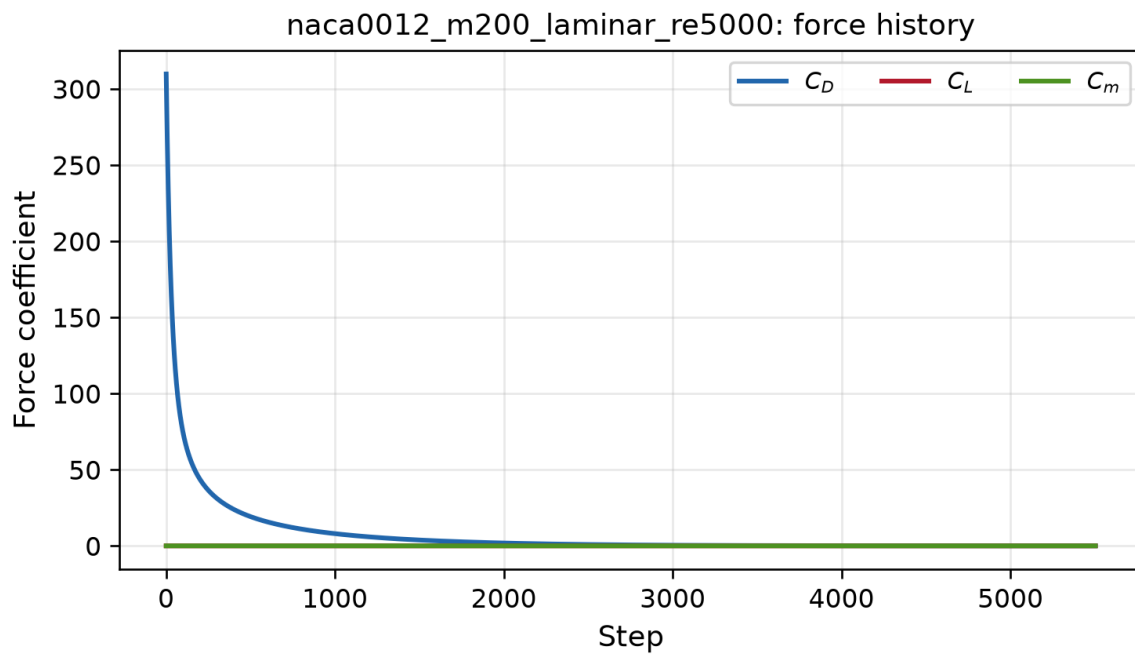


Figure 37: Aerodynamic force-coefficient history.

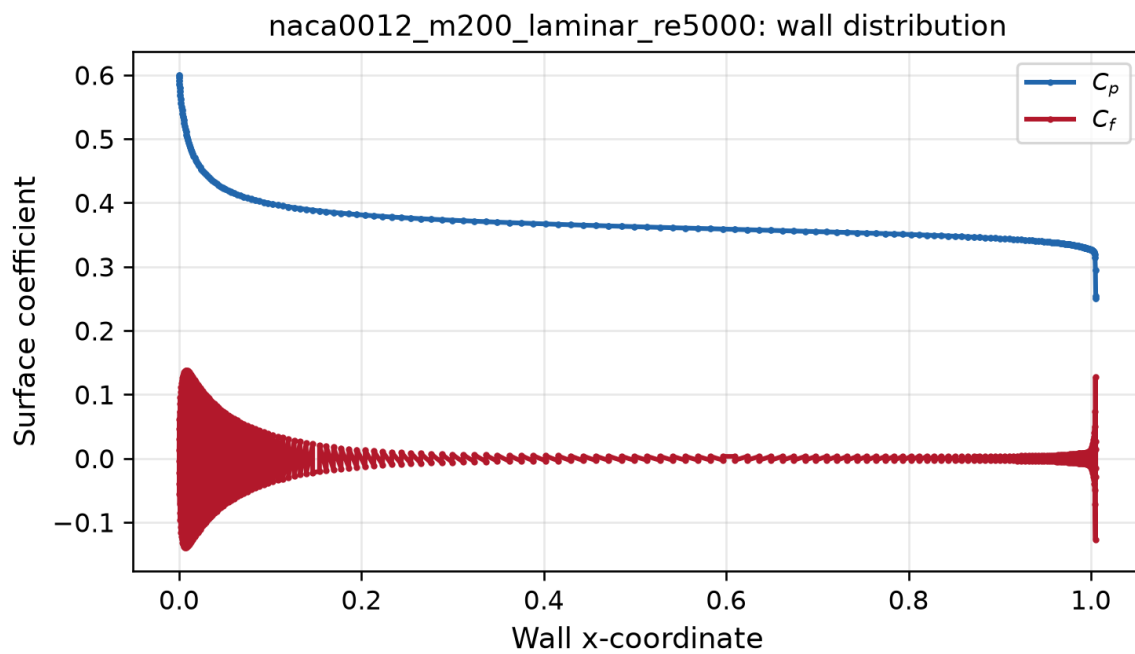


Figure 38: Wall pressure coefficient and skin-friction coefficient.

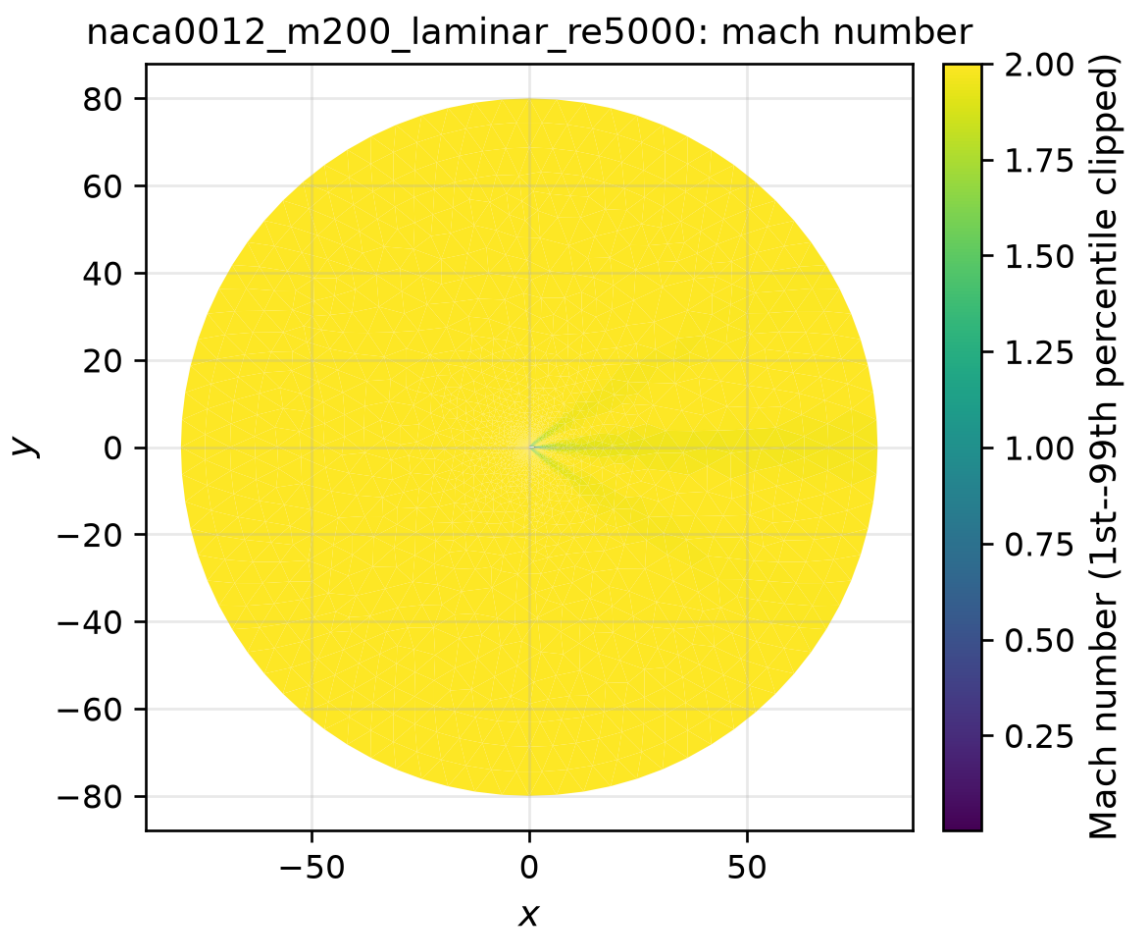


Figure 39: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

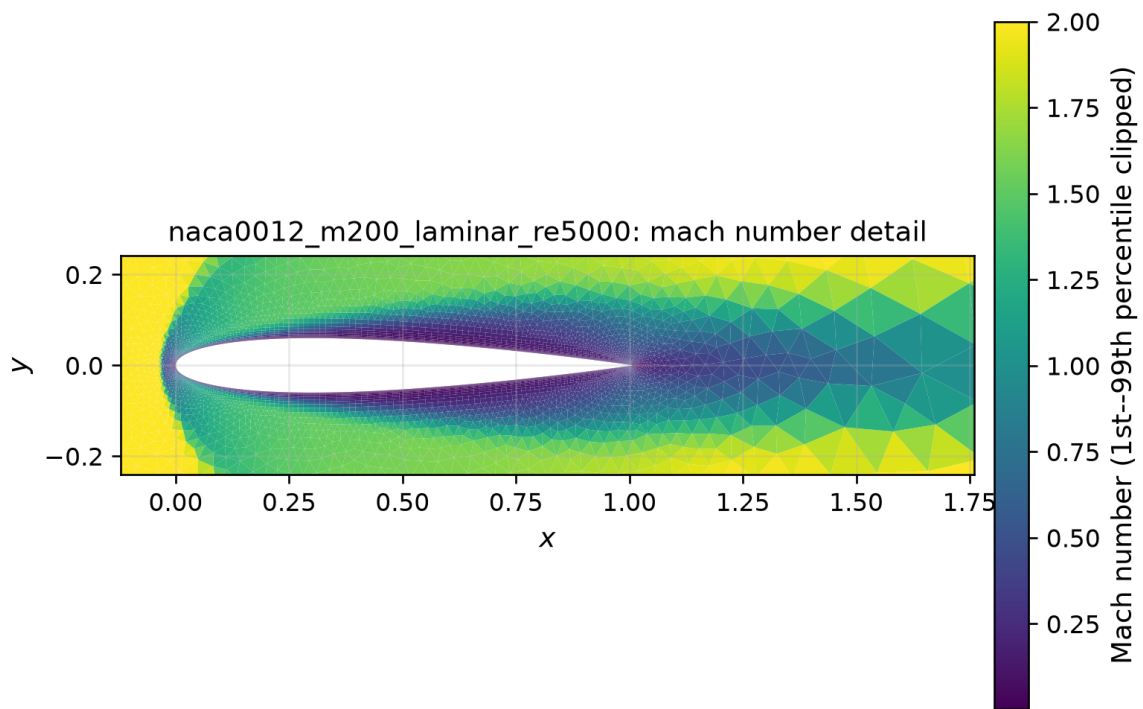


Figure 40: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

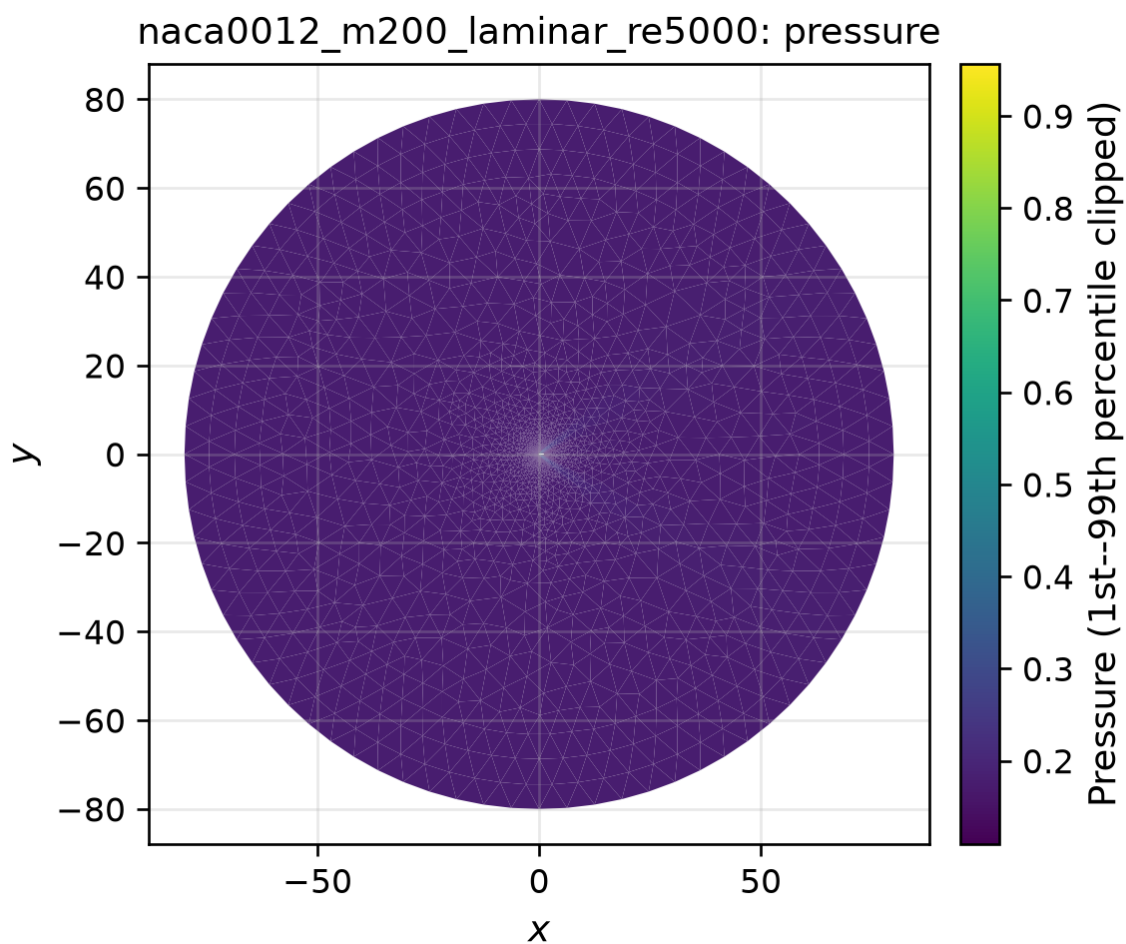


Figure 41: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

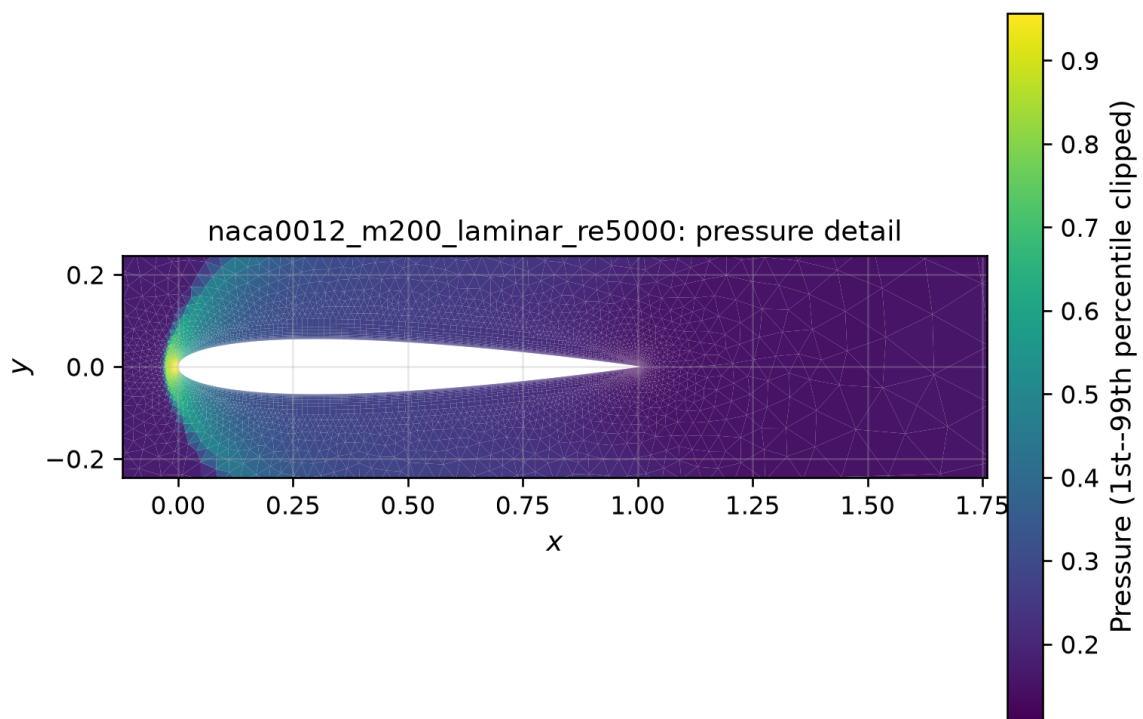


Figure 42: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

10.7 cylinder_m010_laminar_re20

Reported status: **converged**. Final steady force evidence gives $C_D = 1.9713$ and $C_L = 0.0042614$ with surface C_p range -0.7227–0.6678. Final cell fields span Mach 0.0006136–0.1193 and pressure 70.53–72.79. At Re20 the terminal force plateau and symmetric near wake are consistent with the expected steady laminar regime; the wake view is plotted evidence rather than a separately fitted wake metric. The plotted quantity is introduced in Figure 43: Global residual-norm history. The plotted quantity is introduced in

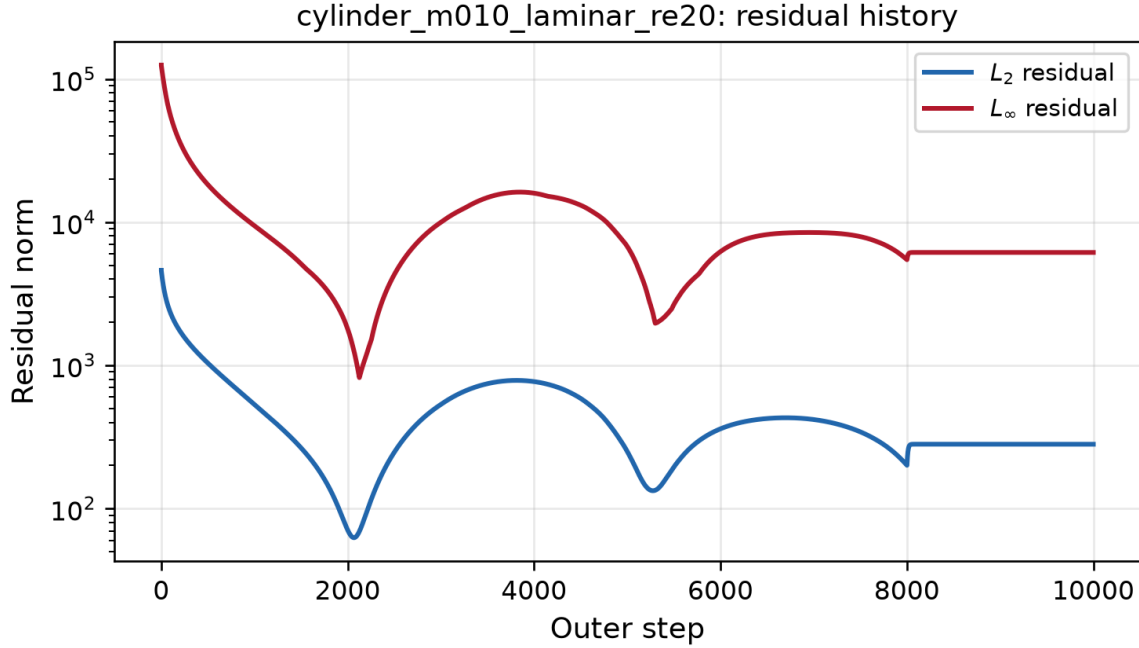


Figure 43: Global residual-norm history.

Figure 44: Aerodynamic force-coefficient history. The plotted quantity is introduced in Figure 45: Wall pressure coefficient and skin-friction coefficient. The plotted quantity is introduced in Figure 46: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 47: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 48: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 49: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

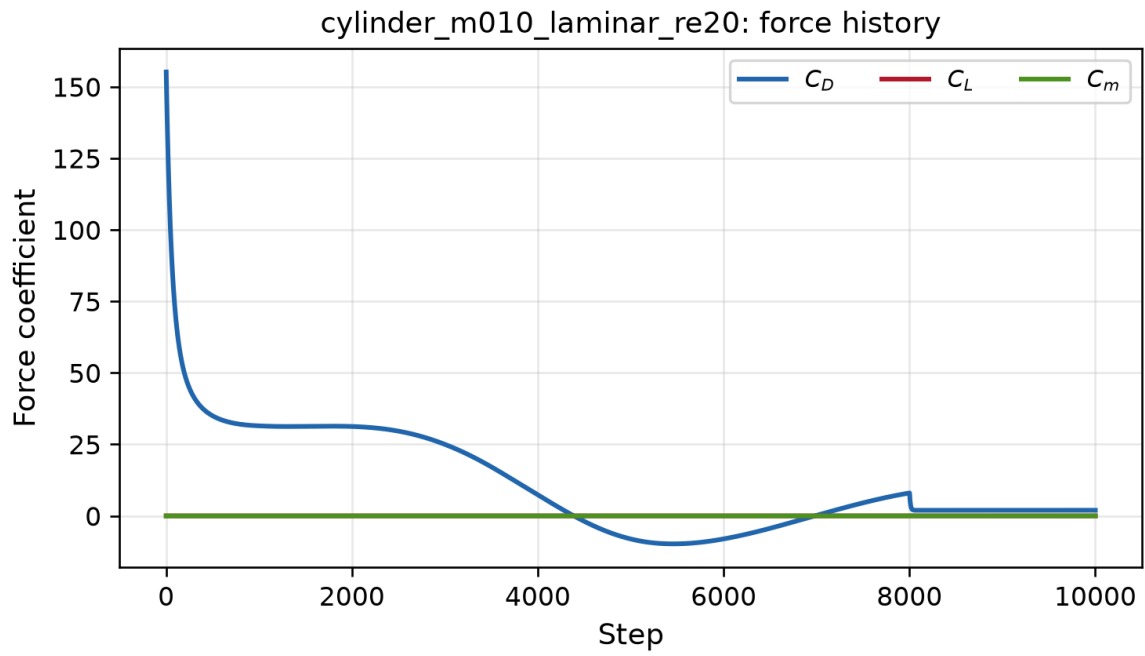


Figure 44: Aerodynamic force-coefficient history.

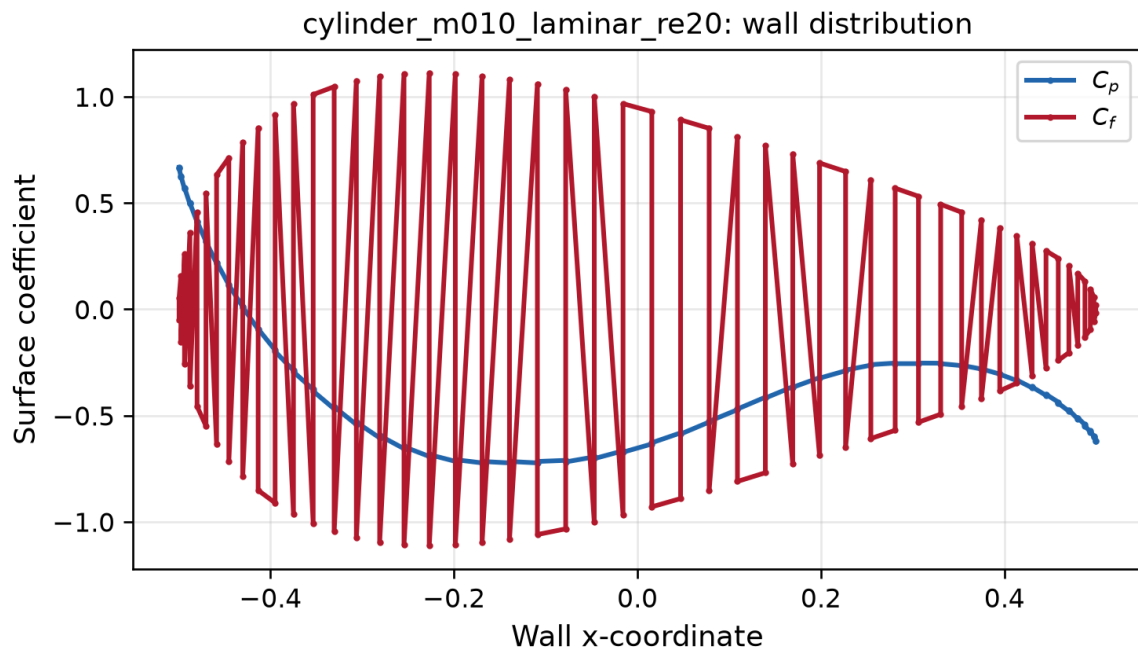


Figure 45: Wall pressure coefficient and skin-friction coefficient.

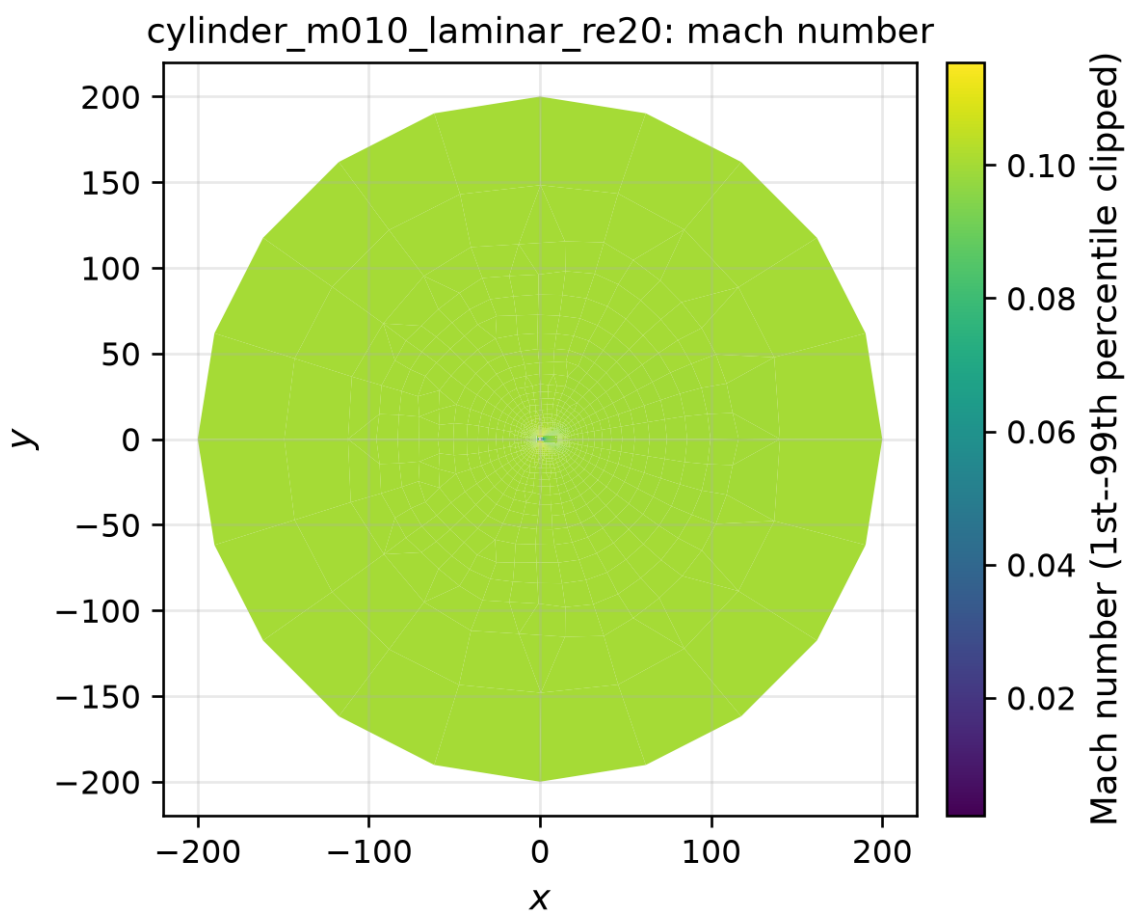


Figure 46: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

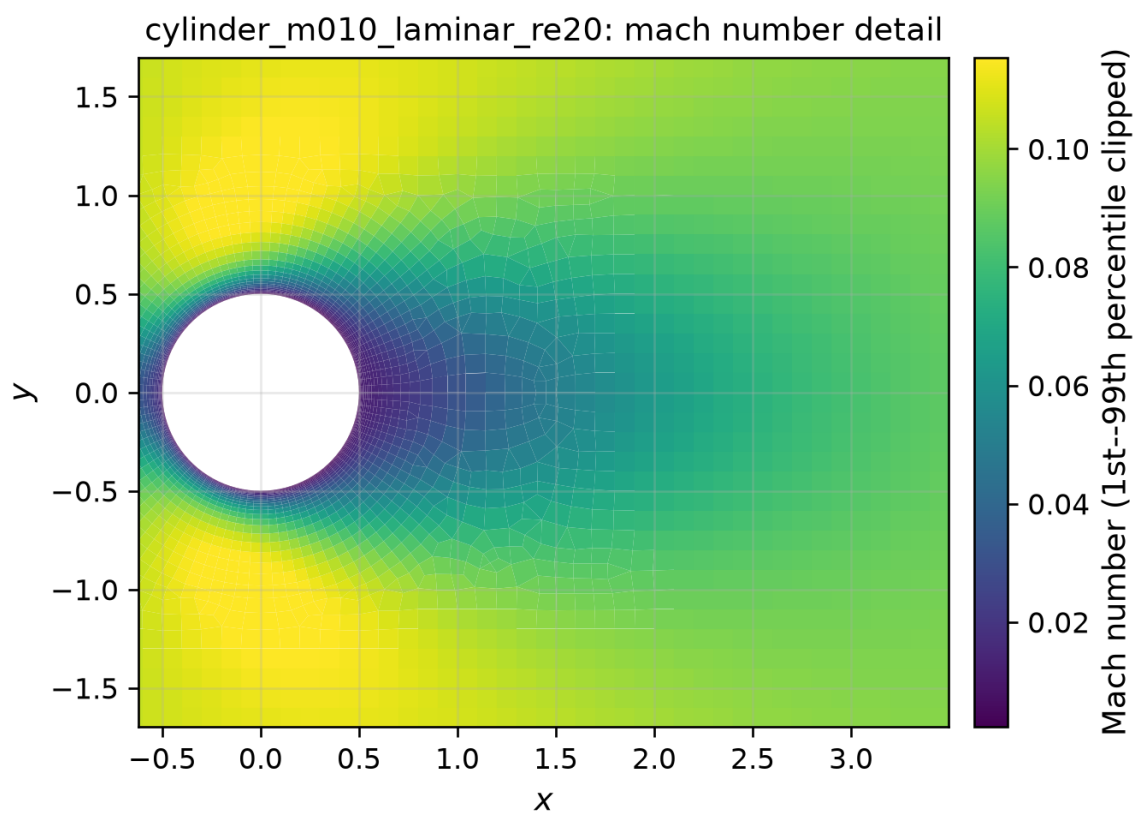


Figure 47: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

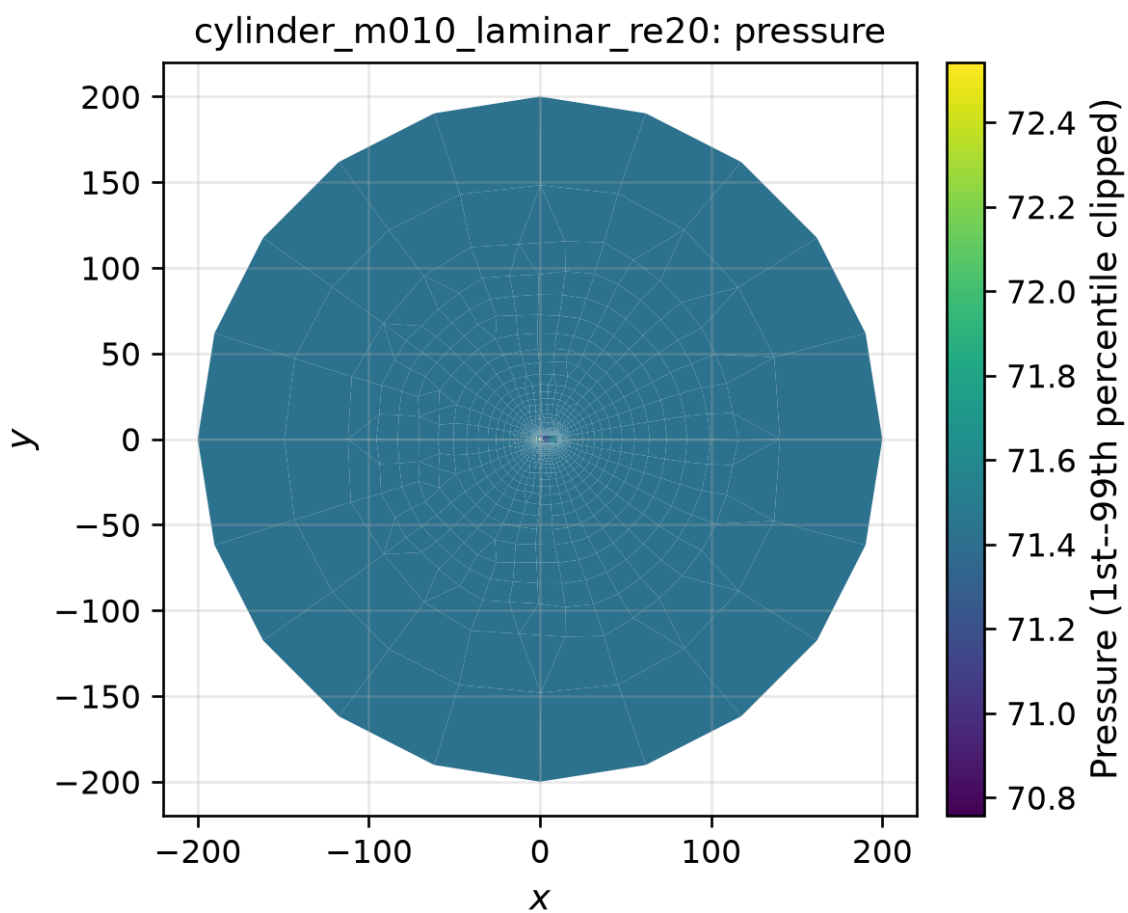


Figure 48: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

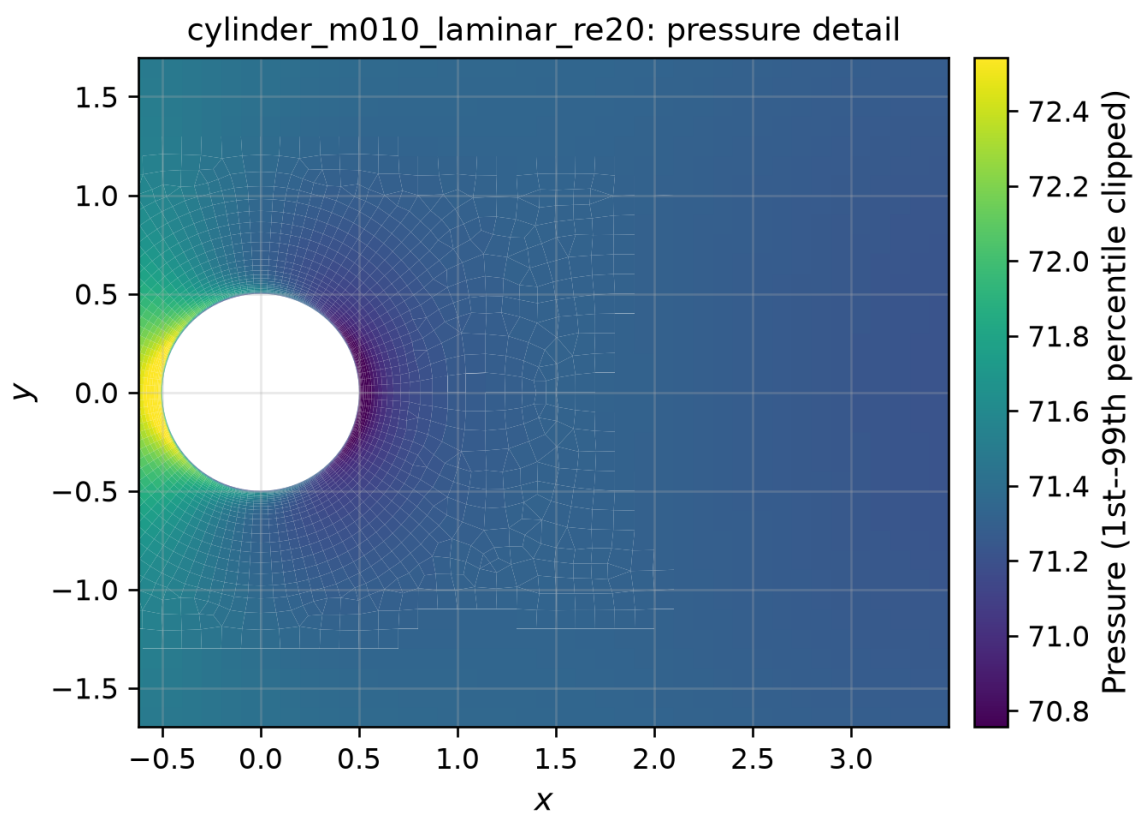


Figure 49: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

10.8 cylinder_m010_laminar_re200

Reported status: **statistically-periodic**. The final-half/post-transient evidence gives mean $C_D = 1.1399$, mean $C_L = -0.00034011$, lift amplitude 0.33817, and dominant frequency $f = 0.16667$ and $St = 0.16667$ over $t = 150.01 - t = 300$. The alternating lift and wake velocity structure demonstrate periodic shedding; 99.6% of physical steps met the inner target. The full numerical spectrum is in `re200_lift_spectrum.csv`. The plotted quantity is introduced in Figure 50: Global residual-norm history. The plotted quantity is introduced in Figure 51: Aerodynamic force-coefficient history. The plotted

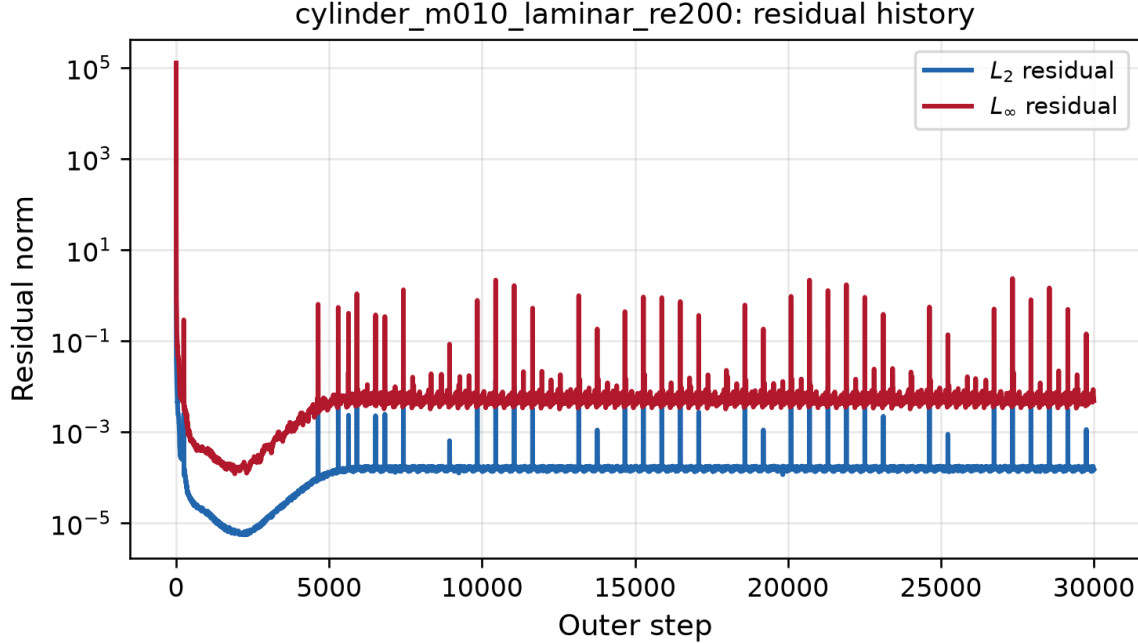


Figure 50: Global residual-norm history.

quantity is introduced in Figure 52: Wall pressure coefficient and skin-friction coefficient. The plotted quantity is introduced in Figure 53: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 54: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 55: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 56: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 57: Cell-polygon velocity magnitude from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles. The plotted quantity is introduced in Figure 58: Post-transient lift spectrum calculated from `forces.csv`; dashed line marks the dominant frequency.

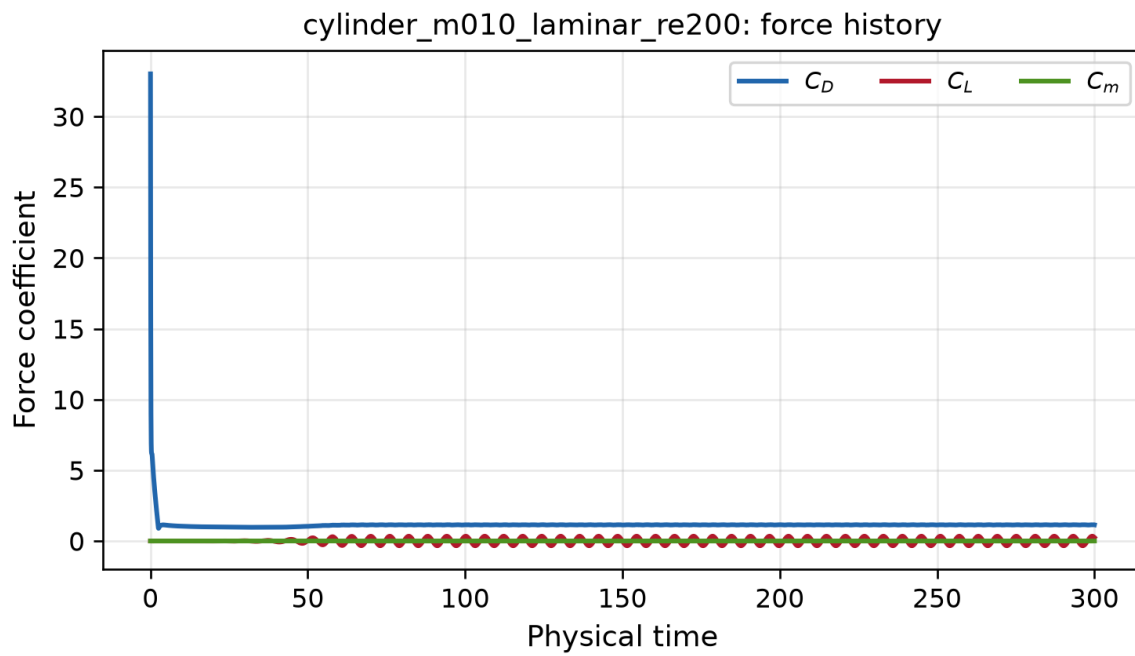


Figure 51: Aerodynamic force-coefficient history.

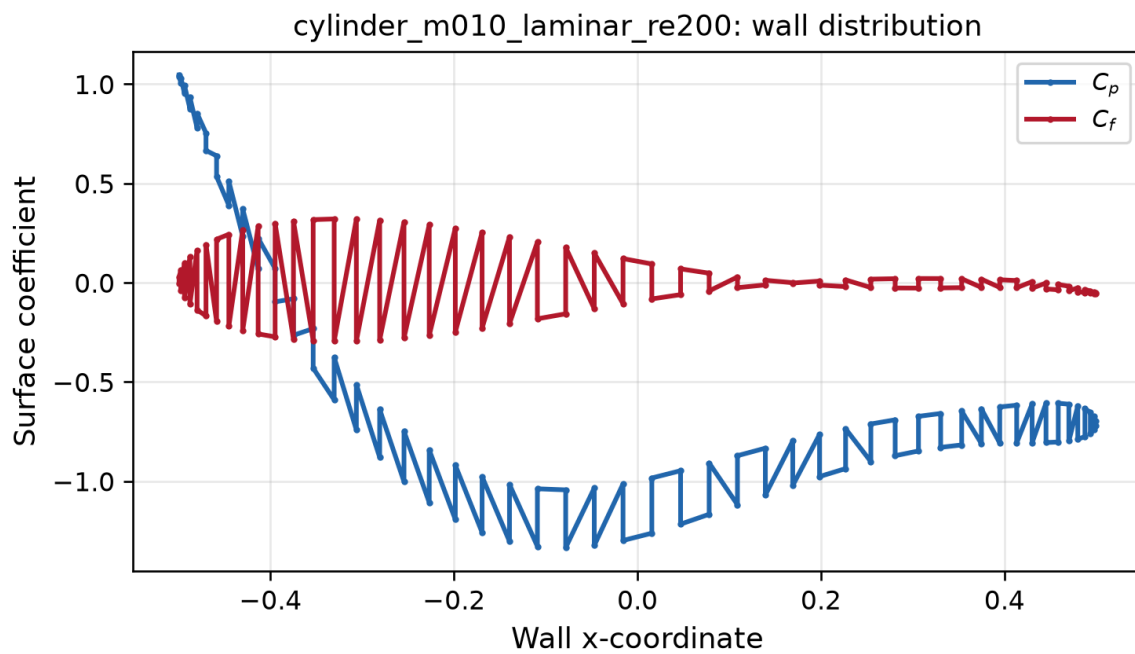


Figure 52: Wall pressure coefficient and skin-friction coefficient.

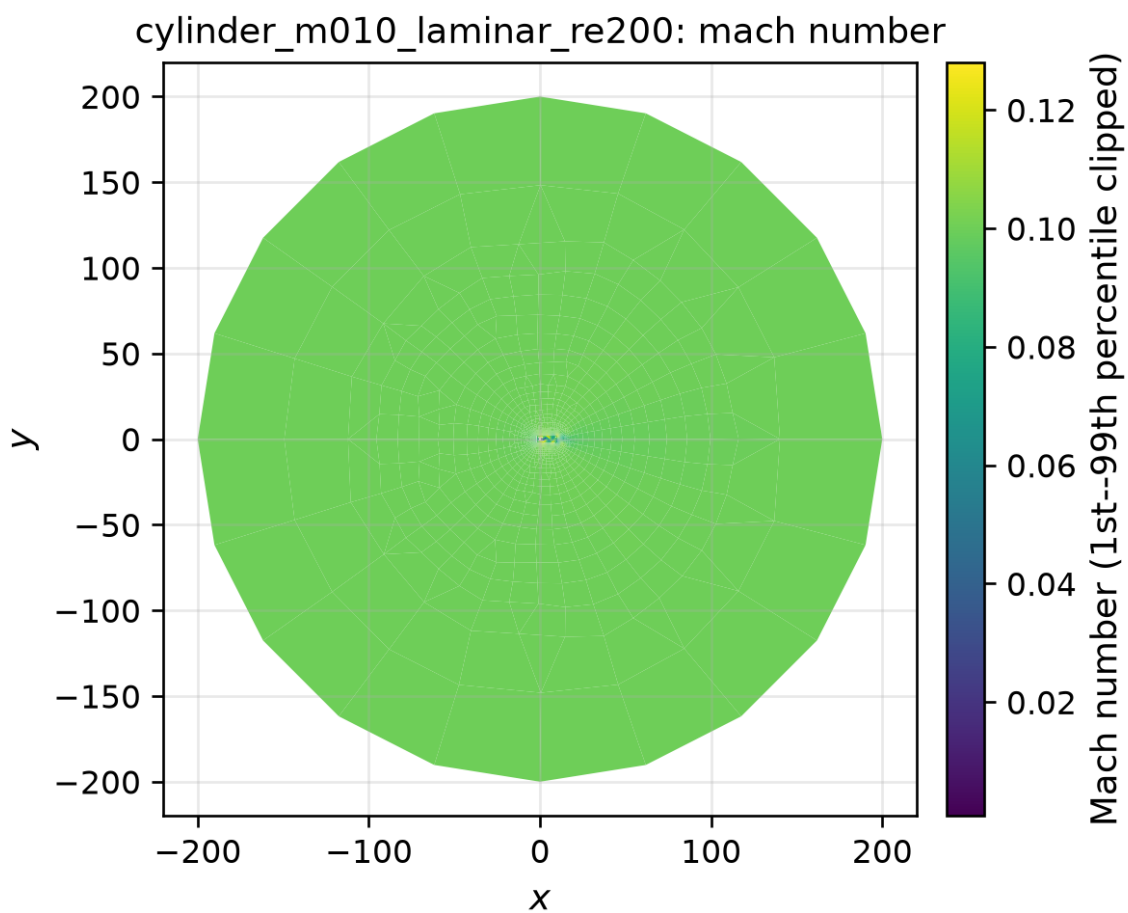


Figure 53: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

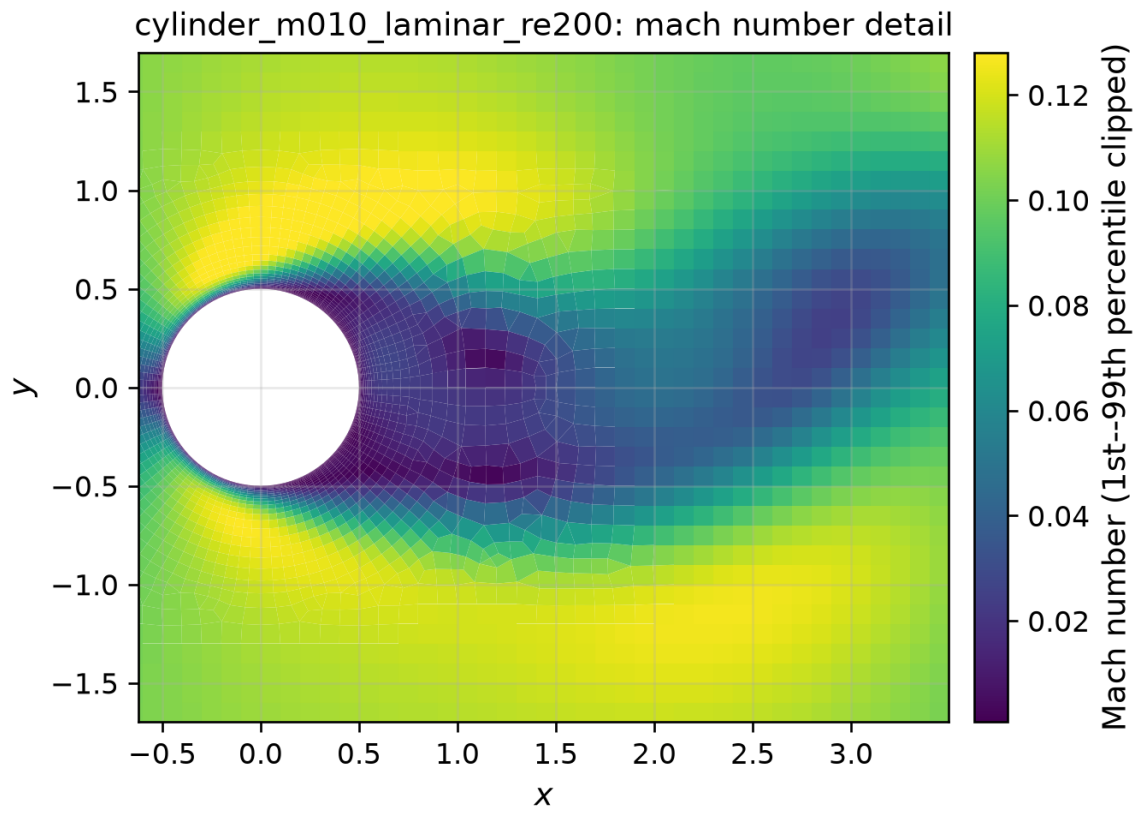


Figure 54: Cell-polygon mach number from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

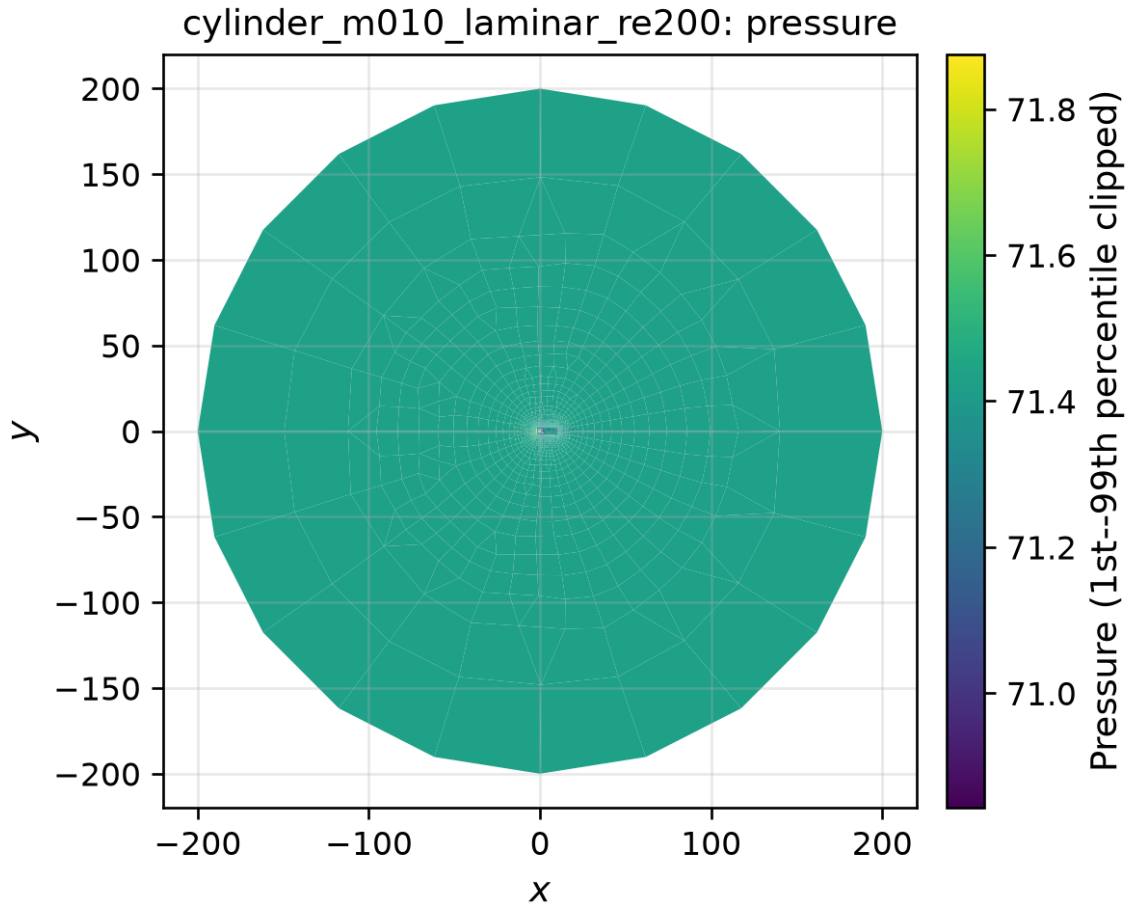


Figure 55: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

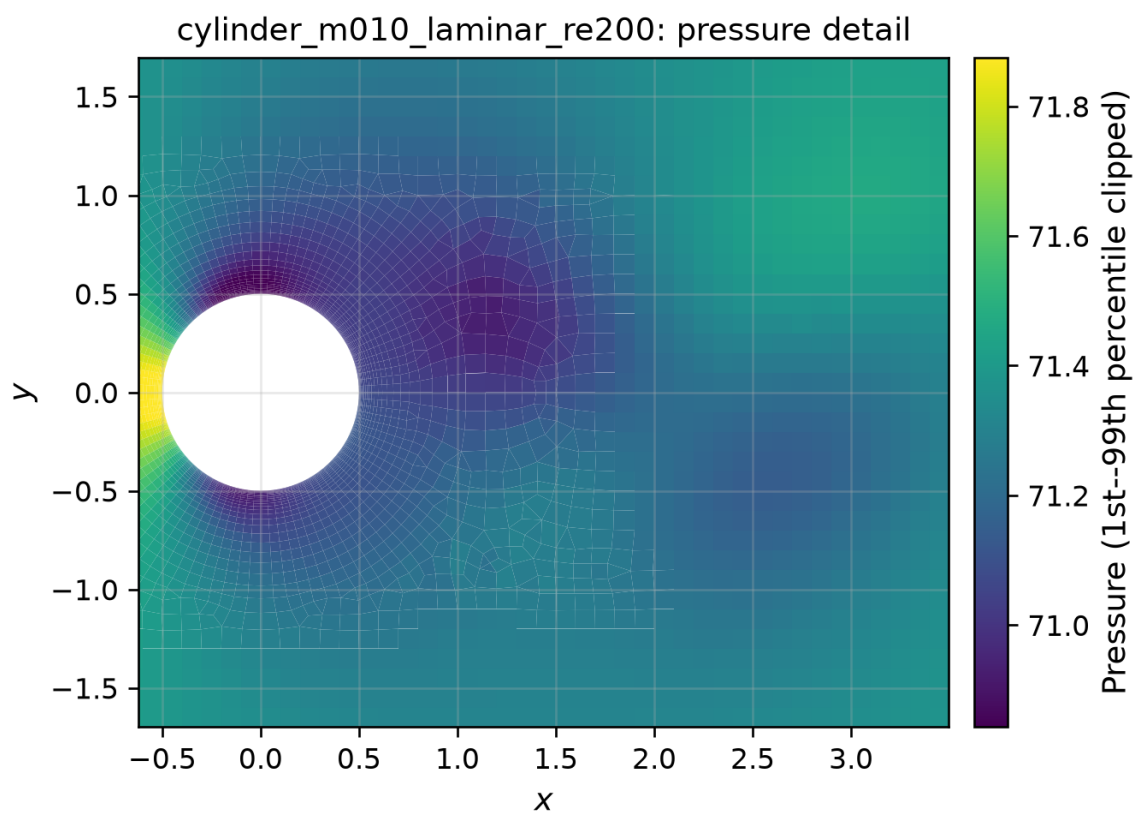


Figure 56: Cell-polygon pressure from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

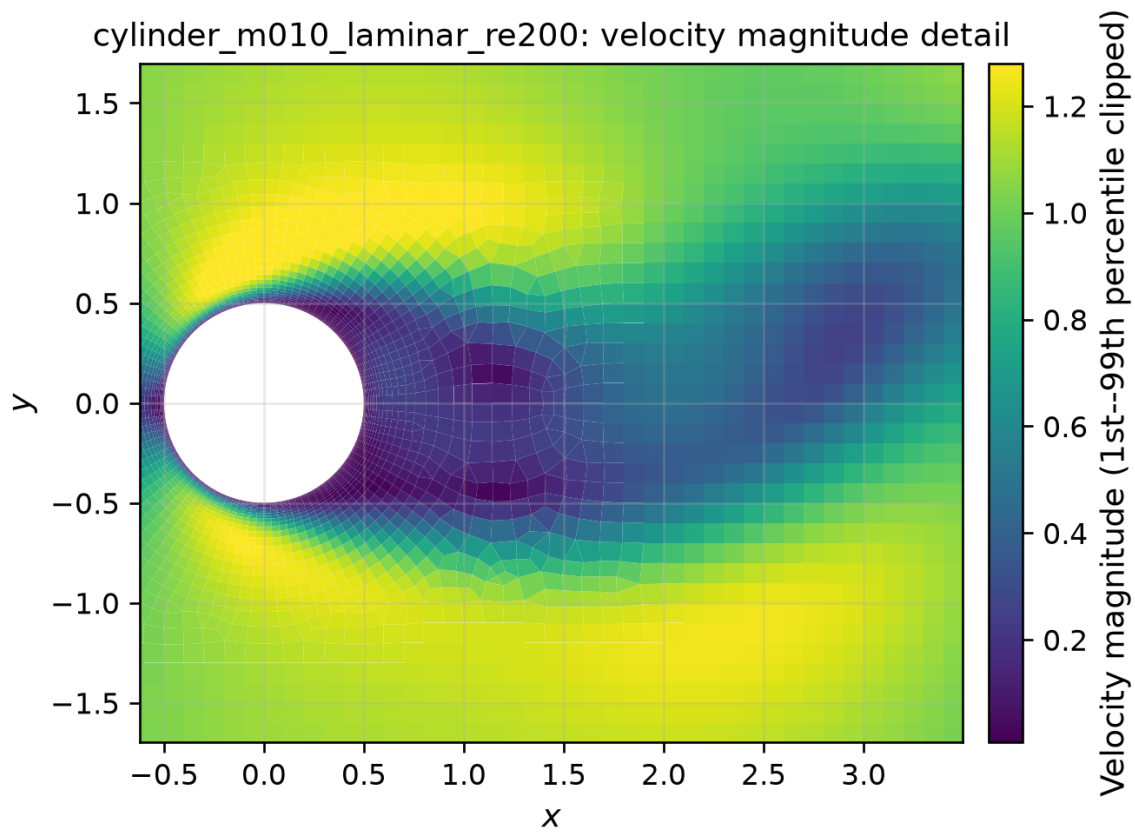


Figure 57: Cell-polygon velocity magnitude from final ASCII VTU field; colour range is clipped to the 1st–99th percentiles.

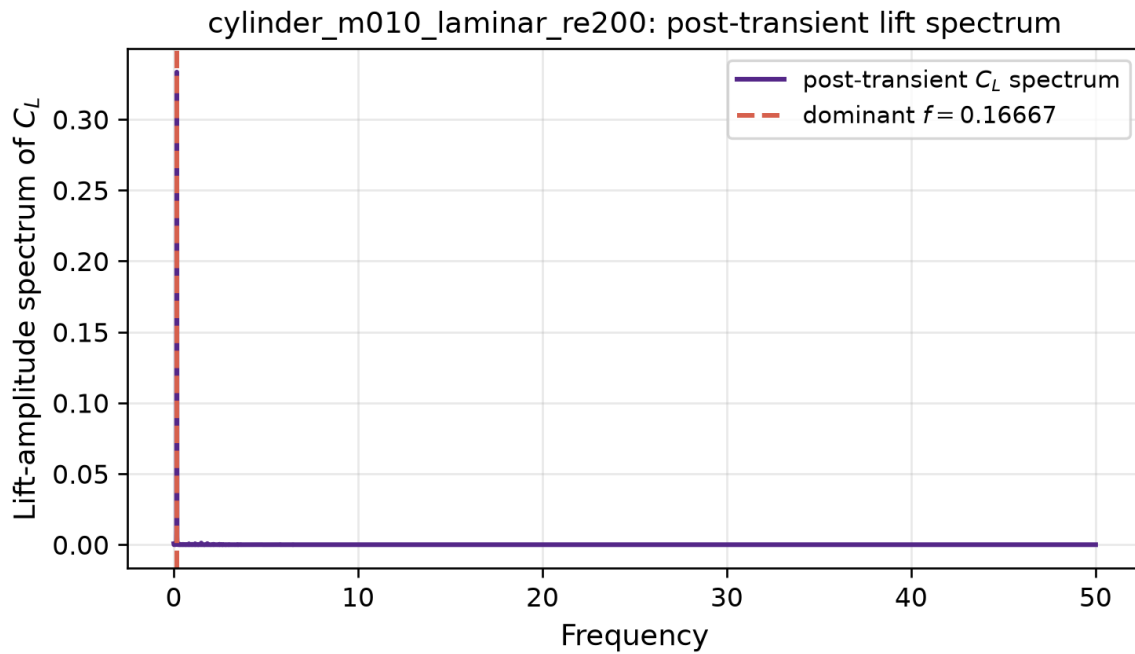


Figure 58: Post-transient lift spectrum calculated from forces.csv; dashed line marks the dominant frequency.

11 Sanity checks

The machine-readable `sanity_checks.json` records positive density/pressure, nontrivial surface pressure, force behavior, final-wall checks, and exact Mach/pressure figure mappings. Viscous walls require maximum reported wall speed at most 10^{-6} and nonzero $|C_f| > 10^{-12}$; inviscid slip walls require normal speed at most 10^{-6} , nonzero tangential speed, and final viscous force at most 10^{-10} . The manifest also confirms that plotted Mach and pressure variables match their file names and source PVTU masters. A failed measured gate forces status `failed`; it is not hidden by the solver’s status claim.

12 Limitations and failure analysis

All required runs completed, so there is no concealed failed production case. Accuracy remains limited by the supplied mesh resolution, constant-viscosity laminar model, scalar-Jacobian LU-SGS approximation, and the deliberately robust but diffusive Rusanov flux; shocks and thin boundary layers are consequently broader than with a less dissipative flux and adapted high-resolution mesh. Reconstruction is first order during its documented startup ramp before reaching the terminal second-order setting. No turbulence, transition, moving mesh, or three-dimensional physics is claimed. The Re200 field output has no cell vorticity array, so its wake figure uses velocity magnitude and cannot expose signed vortex cores as directly. Field colourbars are cellwise 1st–99th percentile clipped, so they suppress extrema and must not be used to infer exact bounds. Rank comparisons are limited to two and eight ranks in one execution environment; they do not establish strong scaling or isolate communication from partitioning and I/O. With more time, the priorities would be a less dissipative shock flux, mesh/adaptation studies, explicit vorticity output, block or Krylov implicit solves, and broader rank scaling.

13 Software design and extension points

Case parsing, CGNS topology, distributed ownership, halo exchange, thermodynamics/fluxes, reconstruction, residual assembly, time integration, and output are separate modules. A three-dimensional extension would add the third momentum component and 3-D face/cell geometry; a general equation of state would replace primitive/conservative conversion and wave-speed closures; RANS or multispecies models would extend the state and flux kernels while retaining partition ownership and typed neighbor exchange. These are design extension points, not implemented capabilities.